

Bipolar Junction Transistors

1. The transistor action
 - 1.1 Operation in the active mode
 - 1.2 Common-base current gain
2. Static characteristics of bipolar transistor
 - 2.1 Carrier distribution in each region
 - 2.2 Ideal transistor currents for active mode operation
 - 2.3 Modes of operation
 - 2.4 Current-voltage characteristics of common-base and common-emitter configuration
 - 2.5 Saturation current and voltage breakdown
3. Frequency Response of Bipolar Transistors
 - 3.1 The Ebers-Moll Model
 - 3.2 Frequency Response

Bipolar Junction Transistors

Bipolar transistor or bipolar junction transistor (BJT) is used extensively in high speed circuits, analog circuits and power applications.

Bipolar devices are semiconductor devices in which both electrons and holes participate in the conduction process. This is in contrast to the unipolar devices, in which only one kind of carrier predominantly participates.

The first bipolar transistor was invented using **two-point contacts on germanium** (by a research team at Bell Laboratories) in 1947.



Co-inventors of the first transistor at Bell Laboratories: Dr. William Shockley (seated); Dr. John Bardeen (left); Dr. Walter H. Brattain. (Courtesy of AT&T Archives.)

Emitter contact wire

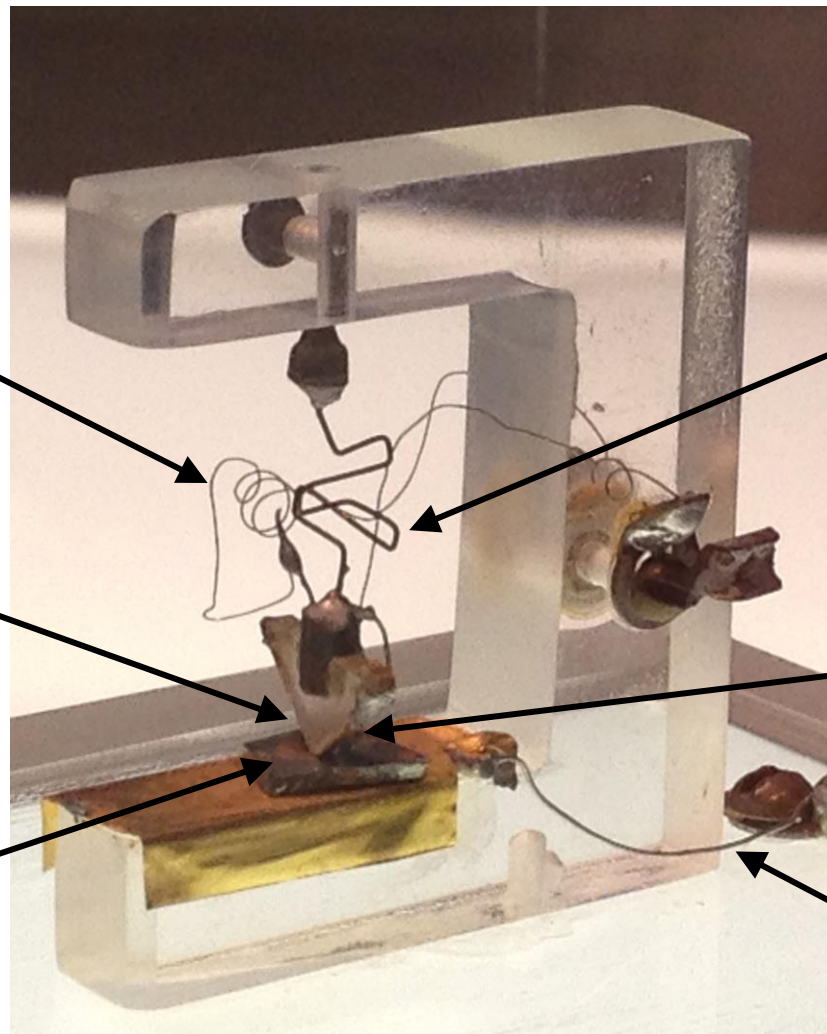
Support wire
(no other use)

Au Emitter (foil)

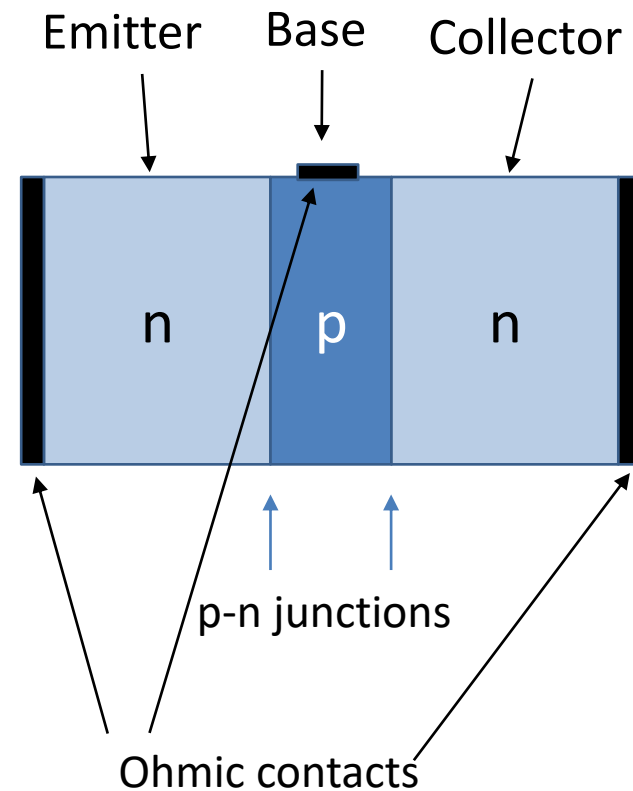
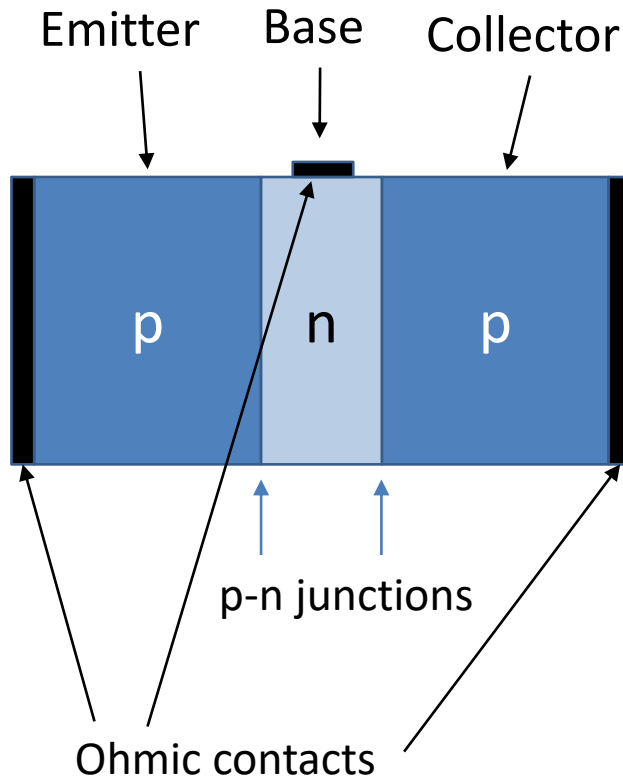
Au Collector (foil)

Ge Base

Base contact wire

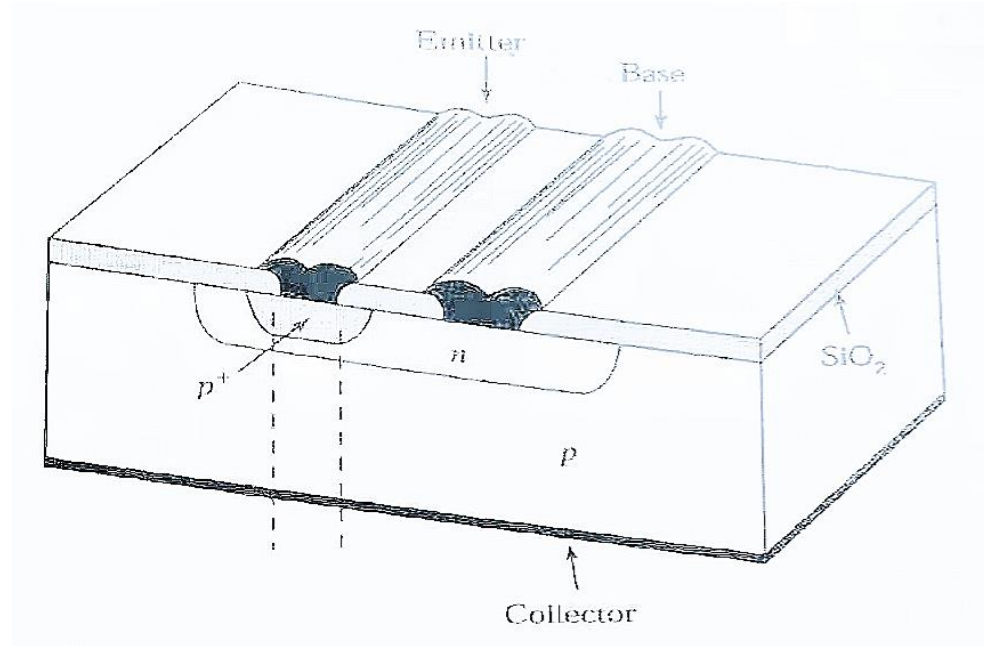


The modern (Si) bipolar transistor consists of **two coupled p - n junctions** in the form of **p - n - p** or **n - p - n** structures.



1. The transistor action

Fig.1 shows the perspective view of a silicon p - n - p bipolar transistor.

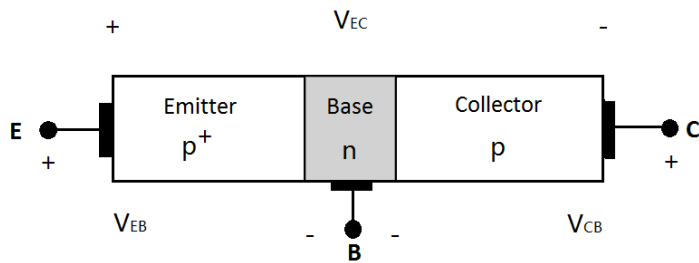


Fabrication:

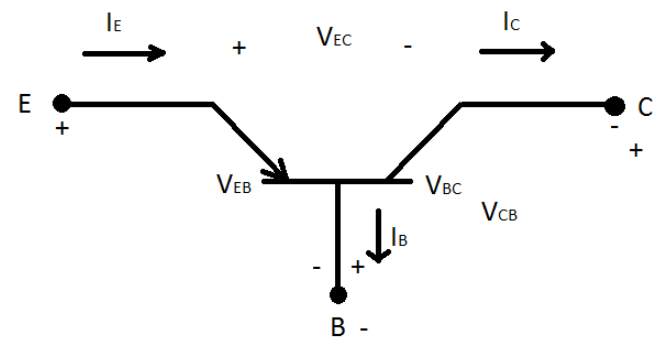
- P -type substrate;
- An n -type region formed by thermally diffusing into the p -type substrate;
- A p^+ -region is formed by diffusing in the n -region;
- Metallic contacts made to the p^+ - and n - regions through the windows opened in the oxide and to the p -region at the bottom.

Fig.2. shows the following:

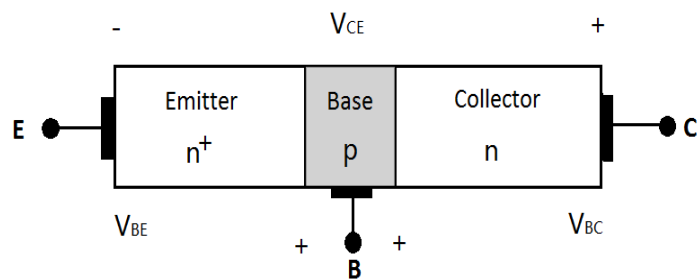
(a) Idealized one-dimensional schematic of a p - n - p bipolar transistor and (b) its circuit symbol. (c) An n - p - n bipolar transistor and (d) its circuit symbol.



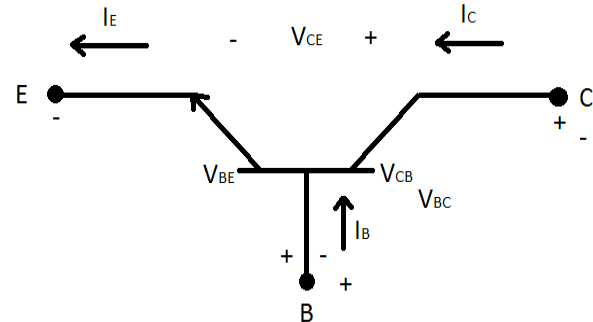
(a)



(b)



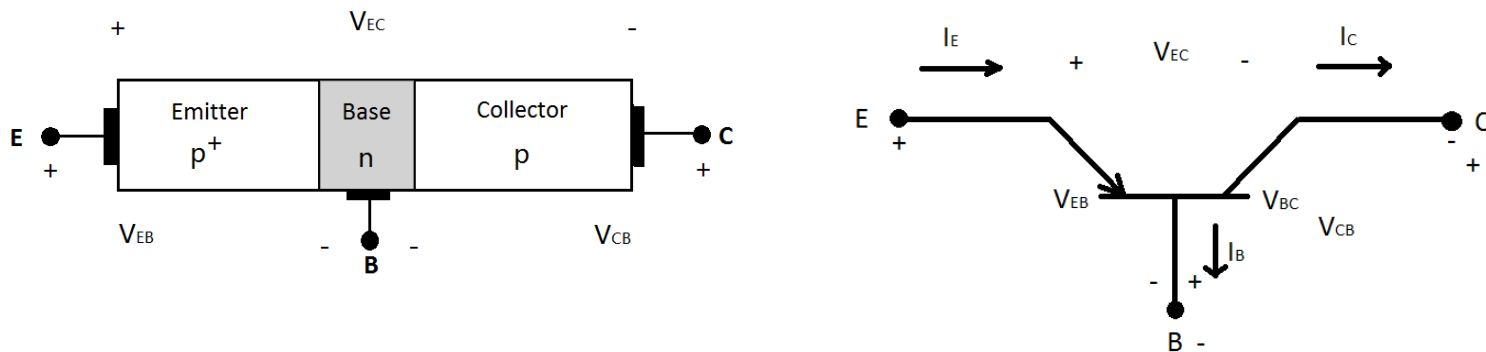
(c)



(d)

The BJT has **three separately doped regions** and **two p - n junctions**.

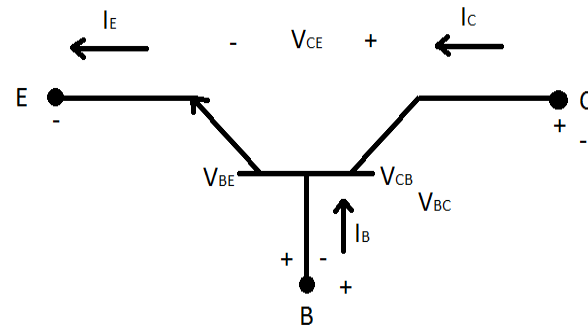
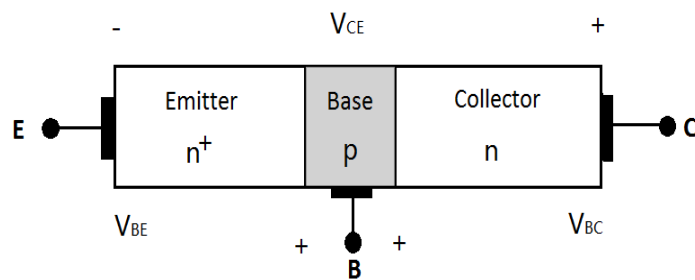
- The heavily doped p^+ -region is called the *emitter (E)*.
- The narrow central region is called the *base (B)*.
- The lightly doped p -region is called the *collector (C)*.



The arrows indicate the directions of current flow under **normal conditions** of the p - n - p BJT (also called *active mode*): that is, the emitter-base junction is forward-biased ($V_{EB} > 0$) and the collector-base junction is reverse-biased ($V_{CB} < 0$).

The complementary structure of the p - n - p transistor is the n - p - n transistor. The current flow and voltage polarity are all reversed.

We shall concentrate on the p - n - p type because the direction of the minority-carrier (hole) flow in the base region is the same as current flow. It provides a more intuitive base for understanding the mechanism of charge transport. Reverse the polarities and conduction types for the n - p - n transistor.



1.1 Operation in the active mode

Fig.3.

(a) A p - n - p transistor with all leads grounded (at thermal equilibrium).

(b) Doping profile of the transistor with abrupt impurity distributions in three regions.

$$N_A(\text{emitter}) > N_D(\text{base}) > N_A(\text{collector}).$$

(c) Electric-field profile.

(d) Energy band diagram at thermal equilibrium.

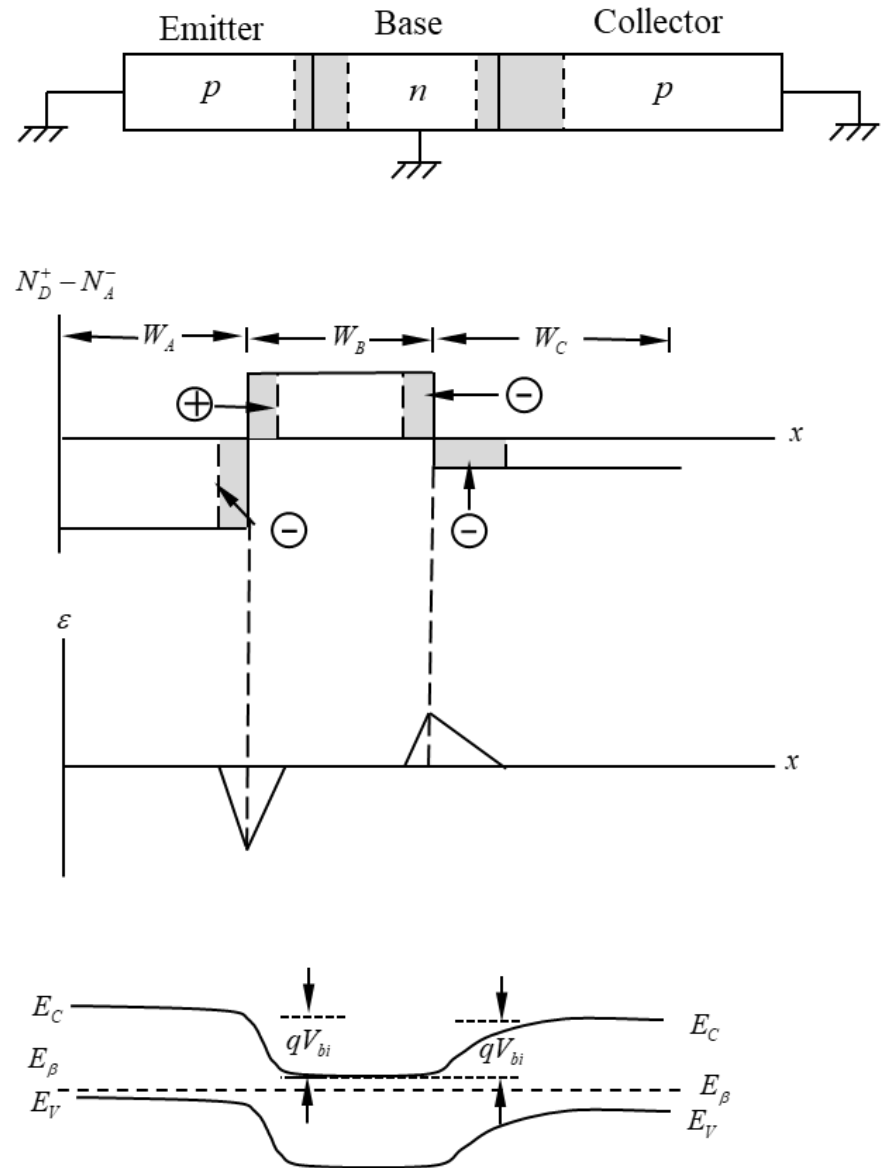


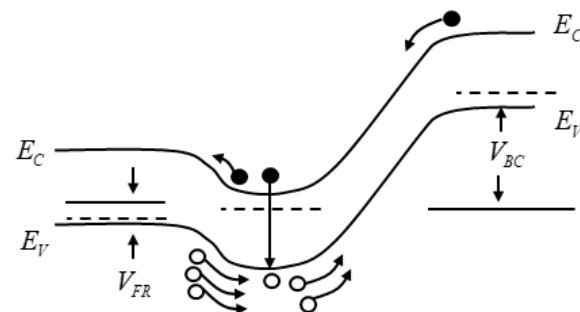
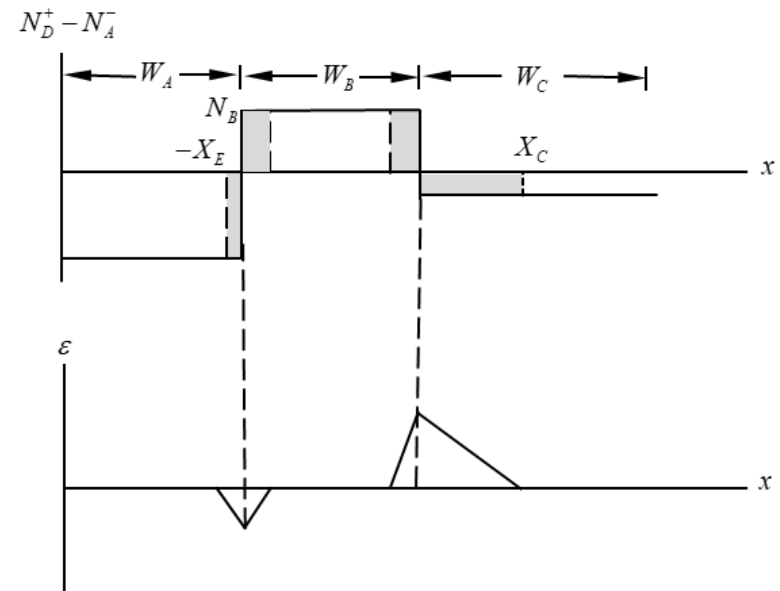
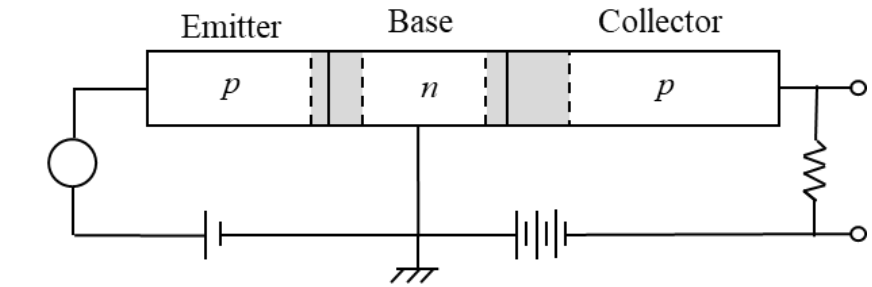
Fig.4.

(a) The transistor shown in Fig.3 under the **active mode of operation**. It is connected as an amplifier with the **common-base configuration**.

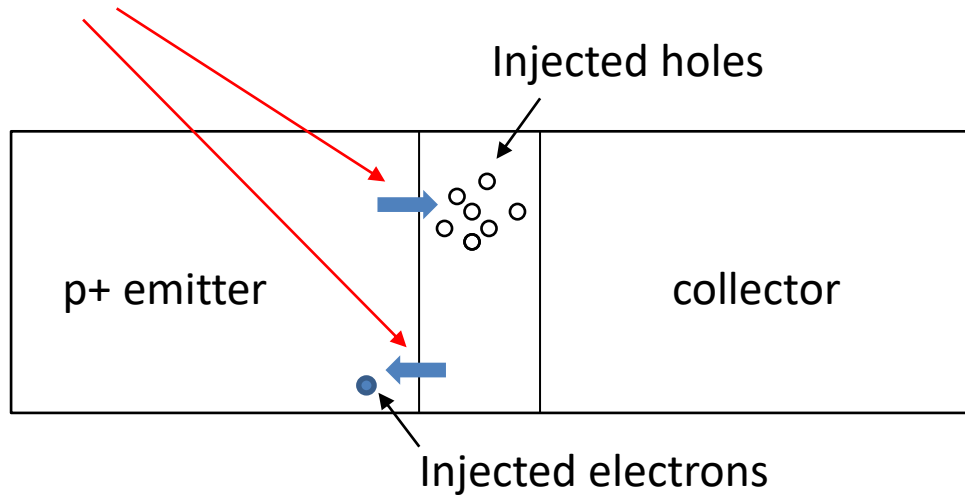
(b) Doping profiles and the depletion regions under biasing conditions. **Note the difference in the depletion layer width.**

(c) Electric-field profile.

(d) Energy band diagram under active mode.

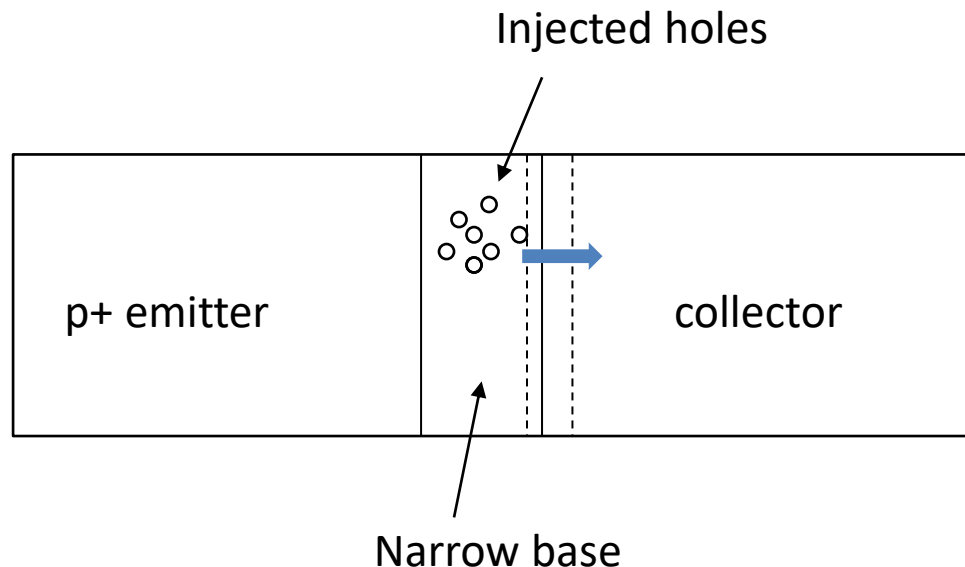


- Since the emitter-base junction is forward-biased, holes will be injected (or emitted) from the p^+ emitter to the base, and the electrons will be injected from the n -type base to the emitter. [Under the ideal-diode condition, there is no generation-recombination current in the depletion region.]
- These two current components constitute the total emitter current.



- The collector-base junction is reverse-biased, and a small reverse saturation current will flow across the junction.

- If the base width is **sufficiently narrow**, the holes injected from the emitter can diffuse through the base to reach the base-collector depletion edge and then “float up” into the collector.
- This transport mechanism gives rise to the terminology of emitter, which *emits* or *injects carriers*, and of collector, *which collects these carriers injected from a nearby junction*.



If most of the injected holes can reach the collector **without recombining** with electrons in the base region, then the collector hole current will be very close to the emitter hole current. ($I_{Ep} \approx I_{Cp}$)

Therefore, carriers injected from a nearby emitter junction can result in a large current flow in a reverse-biased collector junction. This is the **transistor action**, and it can be realized only when the two junctions are physically close enough to interact in the manner described. The two p-n junctions are called the interacting p-n junctions.

1.2 Common-base current gain

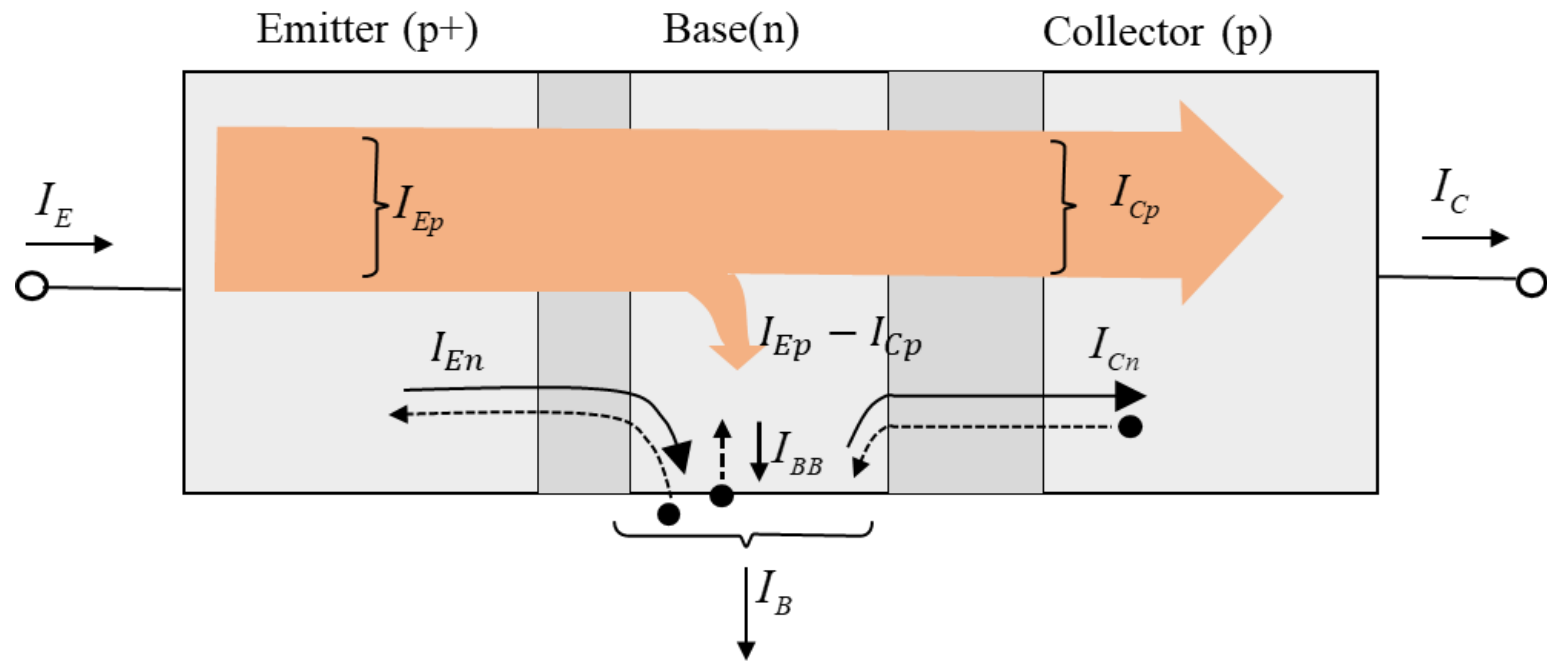
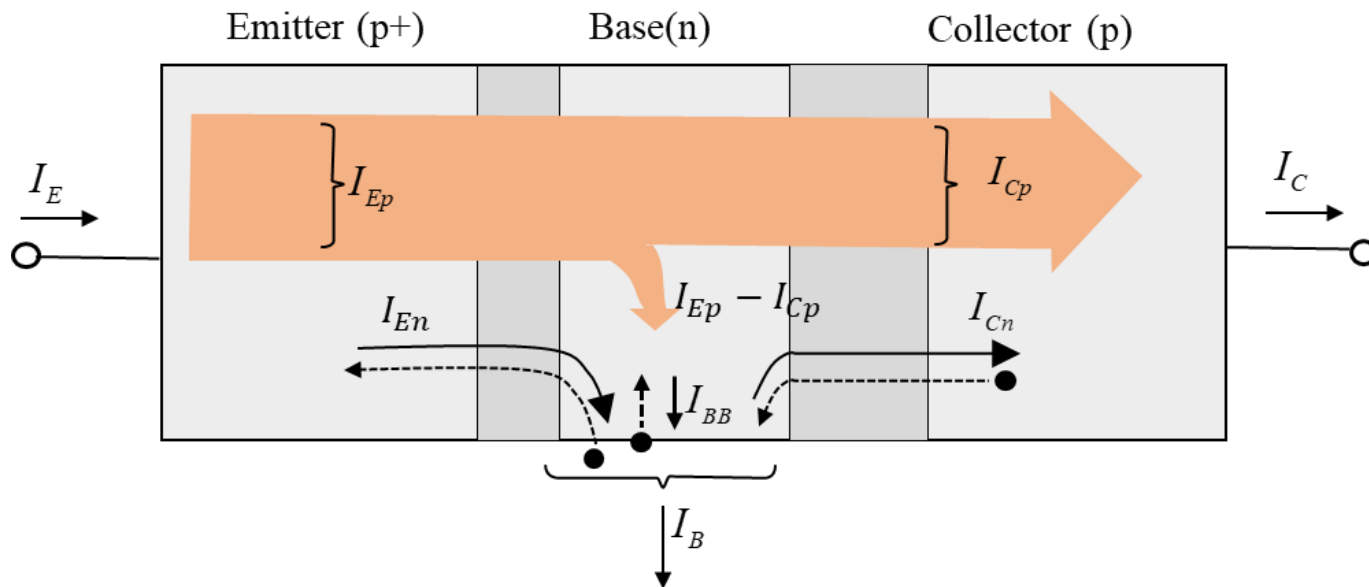


Fig. 5 Various current components in a p-n-p transistor under the active mode of operation. The electron flow is in the opposite direction to the electron current.

I_{Ep} : the current of holes injected from the emitter, which is the largest current component in the well-designed transistor.

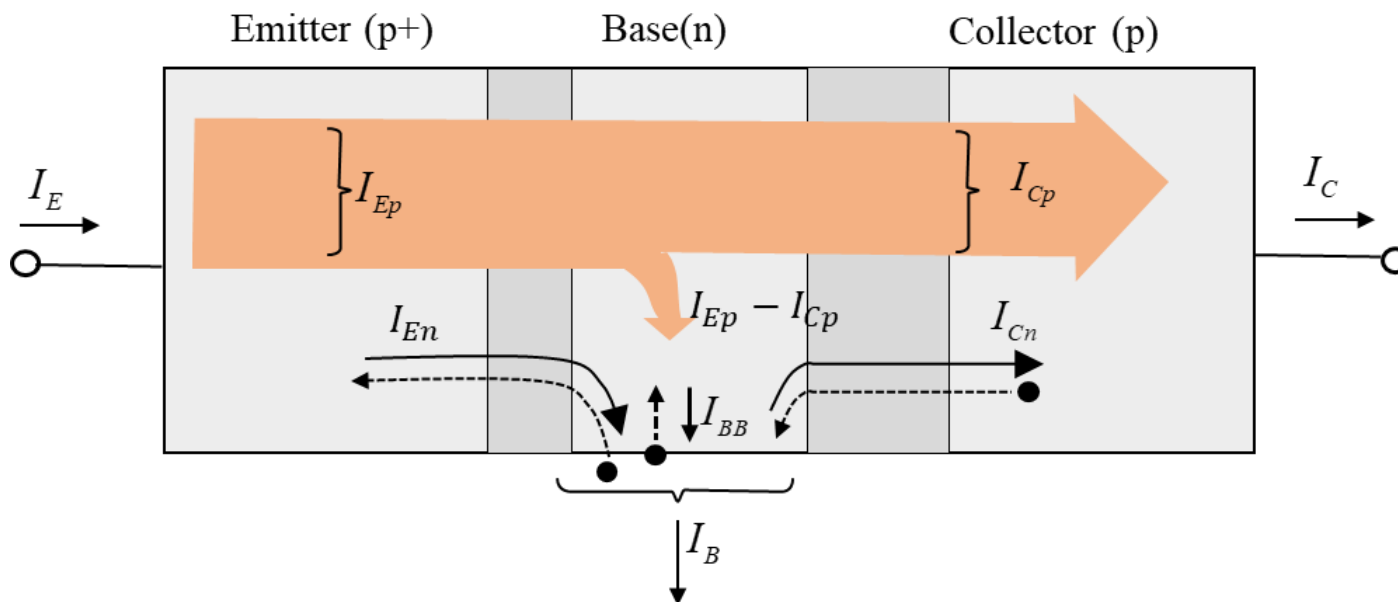
I_{Cp} : the current of holes which reach the collector junction after injected from the emitter.

I_{En} : the current arising from the electrons being injected from the base to the emitter (which is not desirable).



I_{BB} : the current due to the electrons that must be supplied by the base to replace the electrons recombined with the injected holes (i.e., $I_{BB} = I_{Ep} - I_{Cp}$).

I_{Cn} : the current corresponding to thermally generated electrons that are near the B - C junction edge and are drifted from the collector to the base (the direction of the electron current is opposite to the direction of the electron flow).

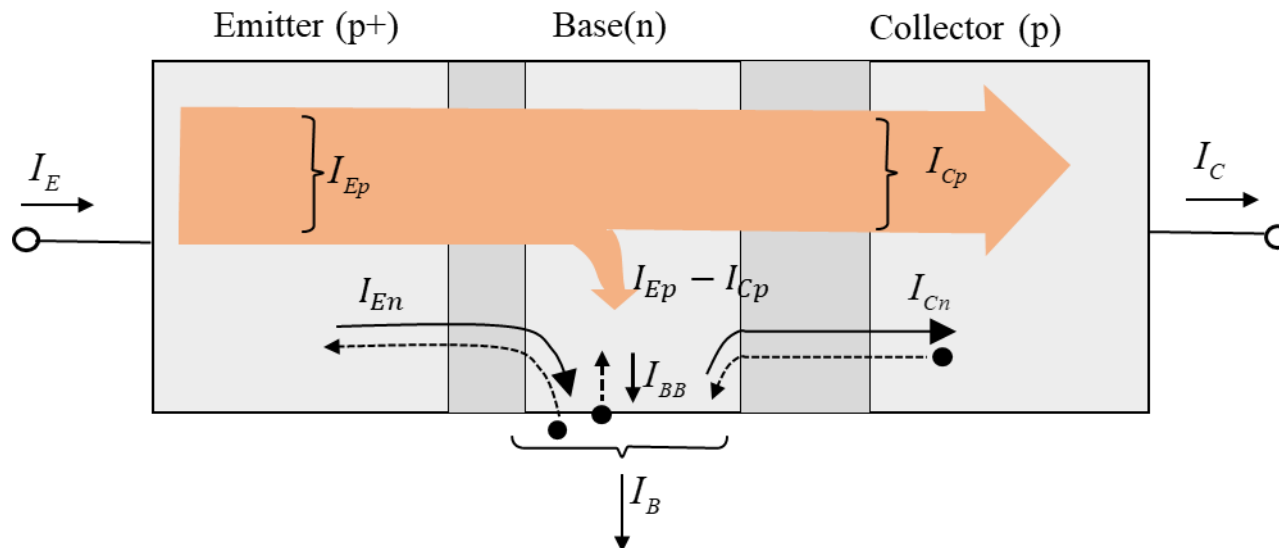


The terminal currents in terms of the various current components described above:

$$I_E = I_{Ep} + I_{En} \quad (1)$$

$$I_C = I_{Cp} + I_{Cn} \quad (2)$$

$$I_B = I_E - I_C = I_{En} + (I_{Ep} - I_{Cp}) - I_{Cn} \quad (3)$$



An important parameter in the characterization of bipolar transistor is the *common-base current gain* α_o , which is defined as

$$\alpha_o = \frac{I_{Cp}}{I_E} = \left[\frac{I_{Ep}}{I_{Ep} + I_{En}} \right] \left[\frac{I_{Cp}}{I_{Ep}} \right] \quad (4) \quad (5)$$

The first term on the right hand side is called the *emitter efficiency* γ , which is a measure of the injected hole current compared to the total emitter current:

$$\gamma = \frac{I_{Ep}}{I_E} = \frac{I_{Ep}}{I_{Ep} + I_{En}} \quad (6)$$

The second term is called the *base transport factor* α_T , which is the ratio of the hole current reaching the collector to the hole current injected from the emitter:

$$\alpha_T = \frac{I_{Cp}}{I_{Ep}} \quad (7)$$

Thus, $\alpha_o = \gamma \alpha_T$ (8)

For a well designed transistor, both γ and α_T approach unity, and α_o is very close to 1.

The collector current in terms of α_o .

$$I_C = I_{Cp} + I_{Cn} = \alpha_o I_E + I_{Cn} \quad (9)$$

Where I_{Cn} corresponds to the collector-base current flowing with the emitter open-circuited ($I_E=0$).

We shall designate I_{Cn} as I_{CBO} , where the first two subscripts (CB) refer to the two terminals between which the current is measured and the third subscript (O) refers to the state of the third terminal with respect to the second. In the present case, I_{CBO} designates the leakage current between the collector and the base with the emitter-base junction open.

$$\text{Therefore,} \quad I_C = \alpha_o I_E + I_{CBO} \quad (10)$$

Example 1

For an ideal p-n-p transistor, the current components are given by $I_{Ep}=3\text{ mA}$, $I_{En}=0.01\text{mA}$, $I_{Cp}=2.99\text{ mA}$ and $I_{Cn}=0.001\text{mA}$. Determine

- (a) the emitter efficiency γ ,
- (b) the base transport factor α_T ,
- (c) the common base current gain α_o and
- (d) I_{CBO} .

Solution:

$$(a) \gamma = \frac{I_{Ep}}{I_E} = \frac{I_{Ep}}{I_{Ep} + I_{En}} = \frac{3}{3+0.01} = 0.9967$$

$$(b) \alpha_T = \frac{I_{Cp}}{I_{Ep}} = \frac{2.99}{3} = 0.9967$$

$$(c) \alpha_o = \gamma \alpha_T = 0.9967 \times 0.9967 = 0.9934$$

$$(d) I_E = I_{Ep} + I_{En} = 3 + 0.01 = 3.01 \text{ mA}$$

$$I_C = I_{Cp} + I_{Cn} = 2.99 + 0.001 = 2.991 \text{ mA}$$

$$I_{CBO} = I_C - \alpha_o I_E = 2.991 - 0.9934 \times 3.01 = 0.87 \text{ } \mu\text{A}$$

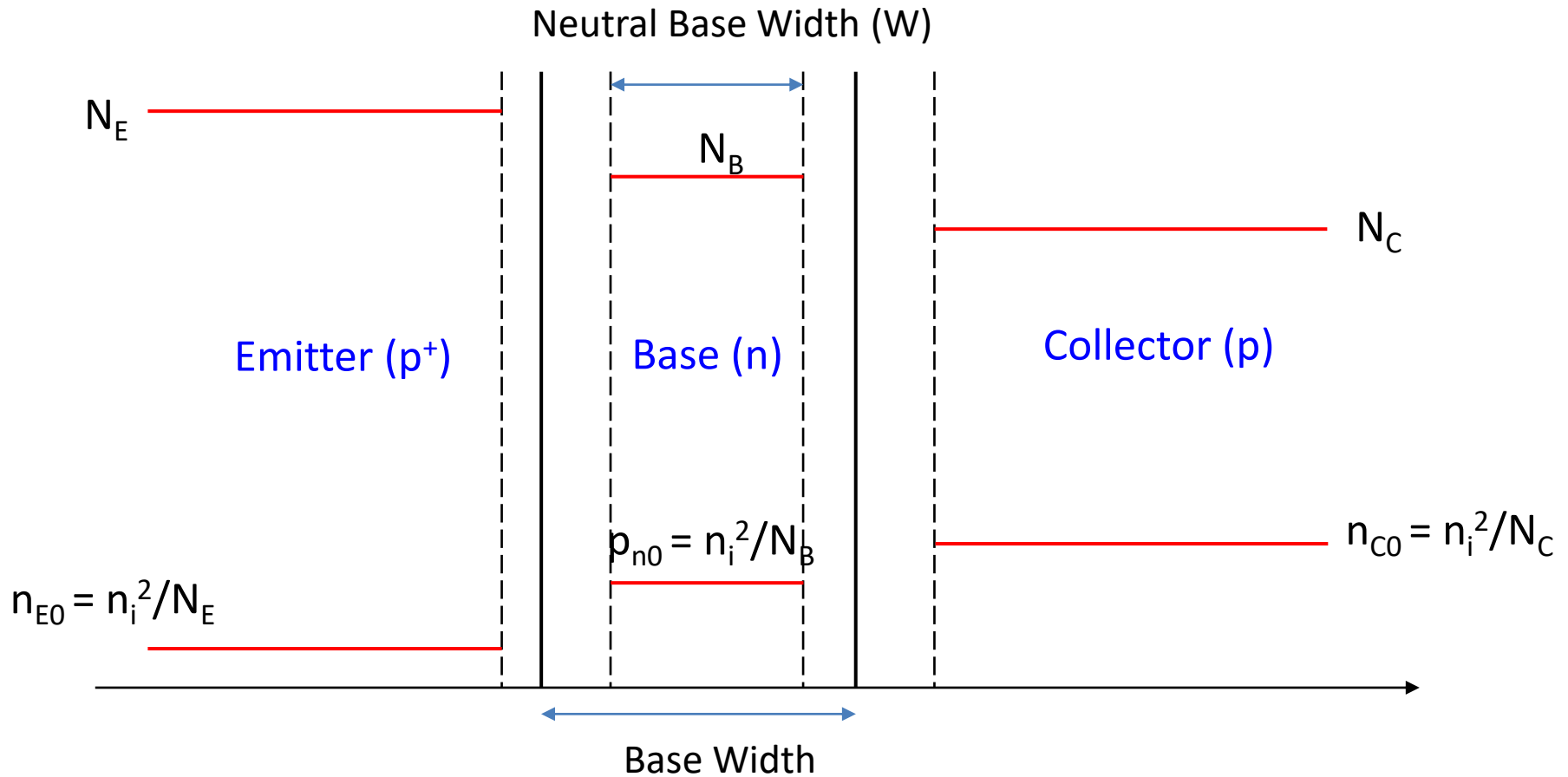
2. Static characteristics of bipolar transistor

2.1 Carrier distribution in each region

To derive the current-voltage expressions for an ideal transistor, we assume the following:

- (1) The device has uniform doping in each region .
- (2) The hole drift current in the base region as well as the collector saturation current is negligible.
- (3) There is low-level injection.
- (4) There are no generation-recombination currents in the depletion regions,
- (5) There are no series resistances in the device.

Terminology of Carrier Concentrations used in BJT



— Carrier concentration

Base region

When the $E-B$ junction is forward biased, the holes are injected to the base region. The minority carrier distribution in the neutral base region can be described by the field-free steady-state continuity equation (similar to the treatment for a junction, as done in p-n junction diode)

$$D_p \left[\frac{d^2 p_n}{dx^2} \right] - \frac{p_n - p_{no}}{\tau_p} = 0 \quad (11)$$

Where D_p and τ_p are the *diffusion coefficient* and *lifetime* of minority carriers, respectively.

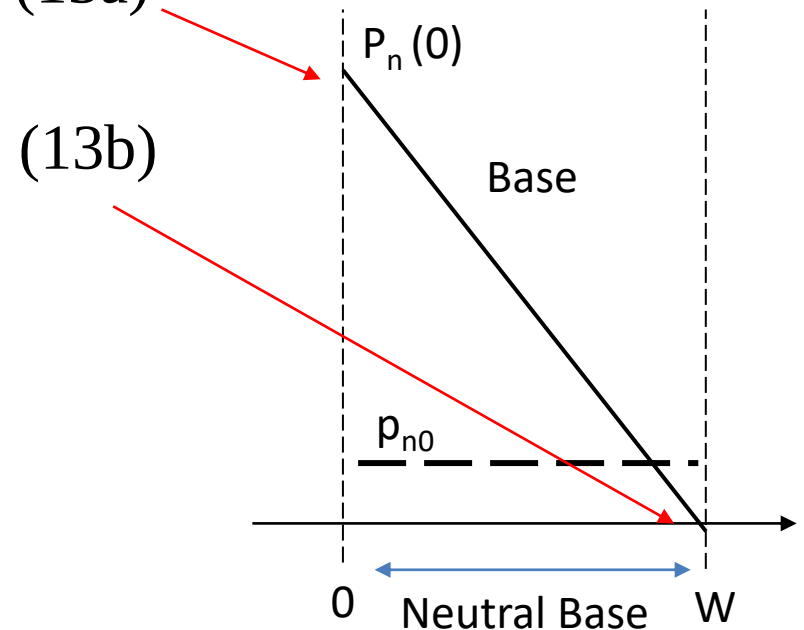
The general solution of Eq.11 is

$$p_n(x) - p_{no} = C_1 e^{x/L_p} + C_2 e^{-x/L_p} \quad (12)$$

Where $L_p = \sqrt{D_p \tau_p}$ is the diffusion length of holes and C_1 and C_2 are constants to be determined by the boundary conditions for the active mode:

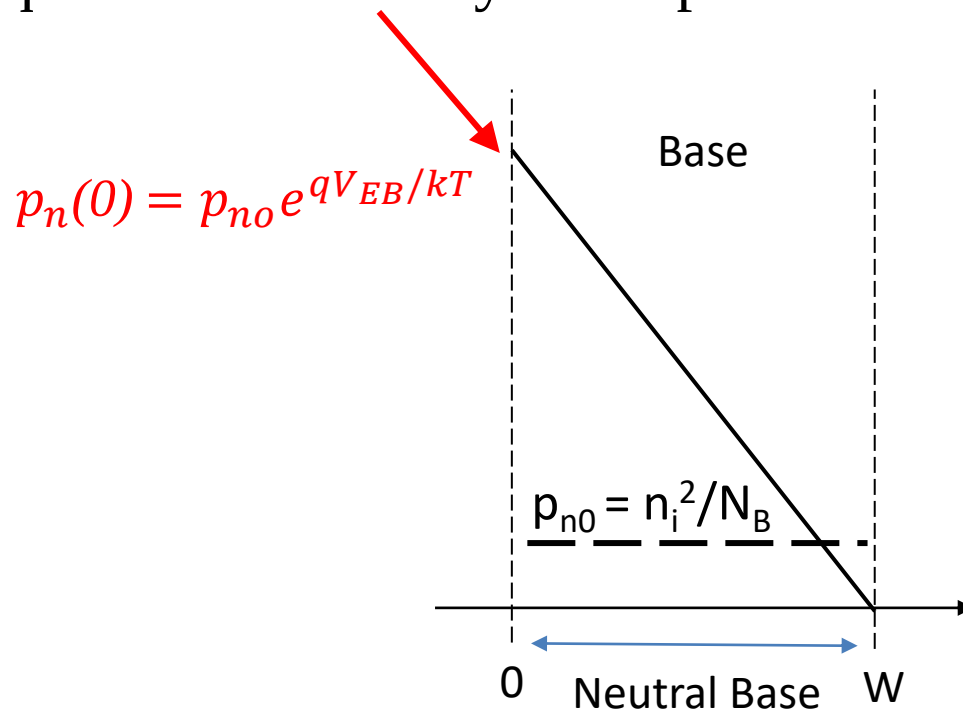
$$\text{And } p_n(0) = p_{no} e^{qV_{EB}/kT} \quad (13a)$$

$$p_n(W) = 0 \quad (13b)$$

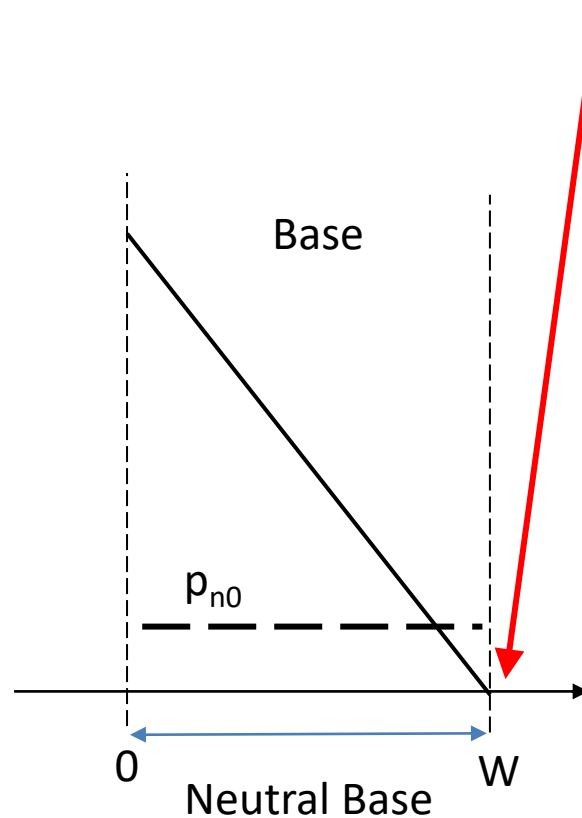


Where p_{no} is the equilibrium minority carrier concentration in the base, given by $p_{no} = n_i^2 / N_B$, and N_B is the uniform donor concentration in the base.

The first boundary condition (Eq.13a) states that under forward bias, the minority hole concentration at the edge of the emitter-base depletion region ($x = 0$) is increased above the equilibrium value by the exponential factor $\exp(qV_{BE}/kT)$.



The second boundary condition (Eq. 13b) states that under reverse bias the minority hole concentration at the edge of the base-collector depletion region ($x = W$) is zero.

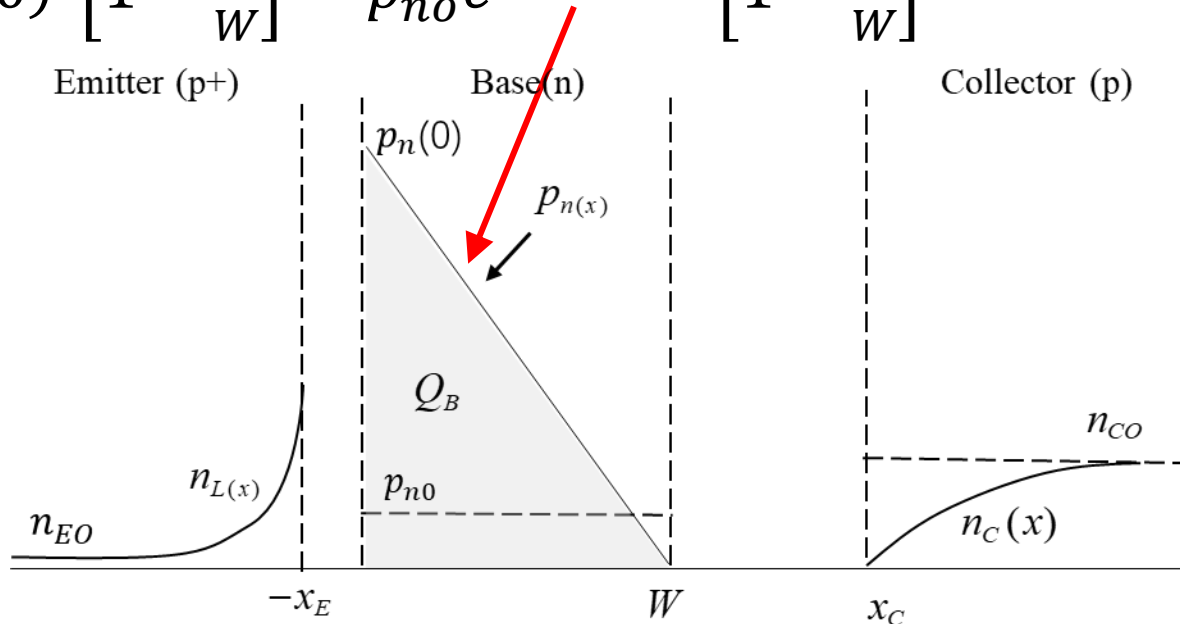


Substituting Eq. 13 into Eq. 12, subjected to the boundary conditions, becomes

$$p_n(x) = p_{no}(e^{qV_{EB}/kT} - 1) \left[\frac{\sinh\left(\frac{W-x}{L_p}\right)}{\sinh\left(\frac{W}{L_p}\right)} \right] + p_{no} \left[1 - \frac{\sinh\left(\frac{x}{L_p}\right)}{\sinh\left(\frac{W}{L_p}\right)} \right] \quad (14)$$

The $\sinh(\Lambda)$ equals approximately to Λ when $\Lambda \ll 1$. When $W/L_p \ll 1$, the distribution can be simplified as

$$p_n(x) = p_n(0) \left[1 - \frac{x}{W} \right] = p_{no} e^{qV_{EB}/kT} \left[1 - \frac{x}{W} \right] \quad (15)$$



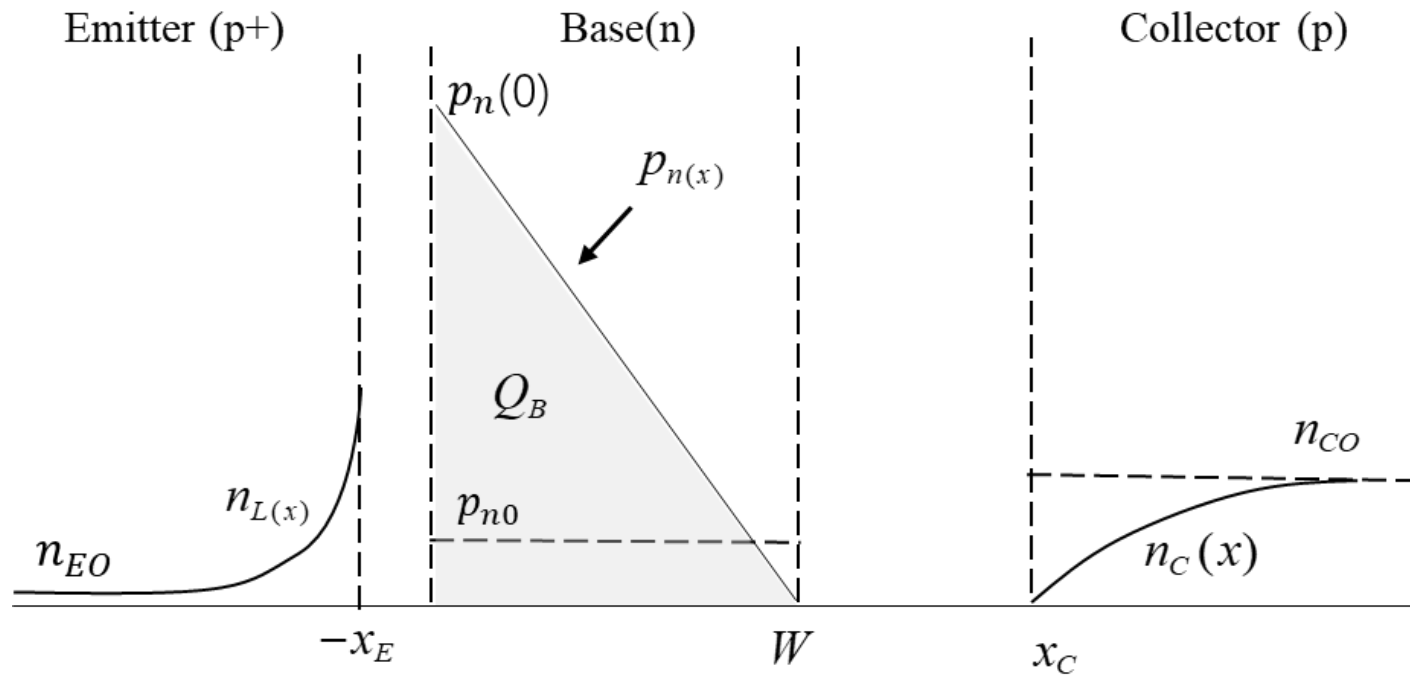


Fig. 6. Minority carrier distribution in various region of a p-n-p transistor under active mode of operation.

Note that as $x/L \ll 1$, by expanding the exponential function into Taylor series and keeping only constant and linear terms in Eq. 12 and solving two algebraic equations with respect to the two constants, one can also show Eq. 15.

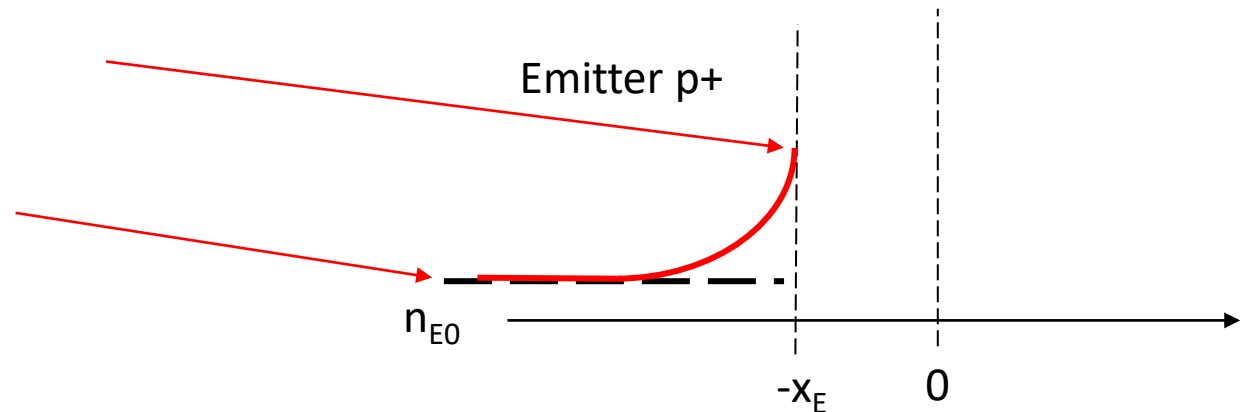
Emitter and collection regions

The minority-carrier distribution in the emitter and collector can be obtained in a manner similar to the one used to obtain the distribution for the base region.

The boundary conditions in **the neutral emitter region** are

$$n_E(x = -x_E) = n_{EO} e^{qV_{EB}/kT} \quad (16)$$

and $n_E(-\infty) = n_{EO}$

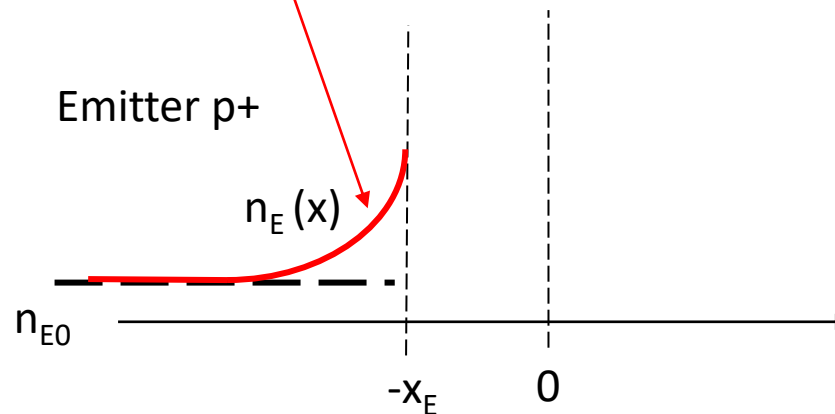


Where n_{EO} is the equilibrium electron concentrations in the emitter.

We assume that the emitter depth is much larger than their corresponding diffusion lengths L_E . Substituting these boundary conditions into an expression similar to Eq. 12 yields

$$n_E(x) = n_{EO} + n_{EO} \left(e^{qV_{EB}/kT} - 1 \right) \exp\left(\frac{x+x_E}{L_E}\right) \quad (17)$$

for $x \leq -x_E$



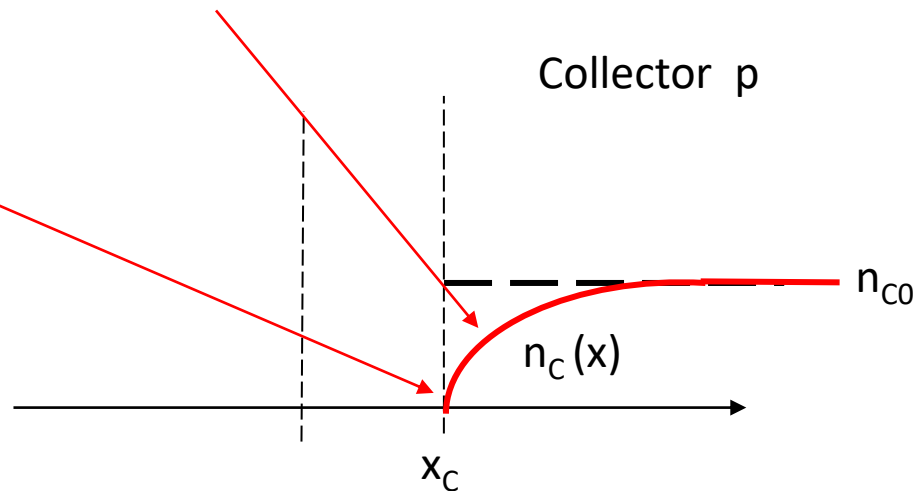
Similarly, **in the collector**, the boundary conditions are

$$n_C(x = x_C) = n_{CO} e^{-q|V_{CB}|/kT} = 0 \quad (18)$$

$$\text{and } n_C(\infty) = n_{CO}$$

The minority carrier concentration is

$$n_C(x) = n_{CO} - n_{CO} \exp\left(-\frac{x-x_C}{L_C}\right) \quad \text{for } x \geq x_C \quad (19)$$

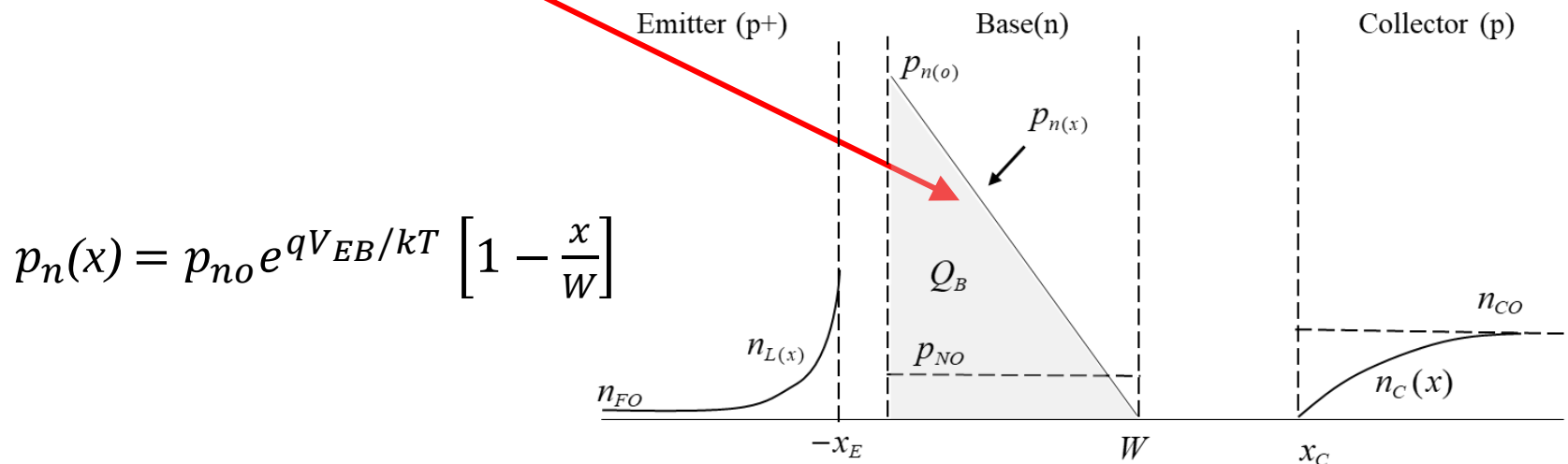


2.2 Ideal transistor currents for active mode operation

Once the minority carrier distributions are known, the various current components shown in Fig.5 can be calculated.

The hole current I_{Ep} , injected from the emitter at $x = 0$, is proportional to the gradient of the minority carrier concentration and is expressed by

$$I_{Ep} = A \left(-q D_p \frac{dp}{dx} \Big|_{x=0} \right) \equiv q A \frac{D_p p_{no}}{W} e^{qV_{EB}/kT} \quad (20)$$

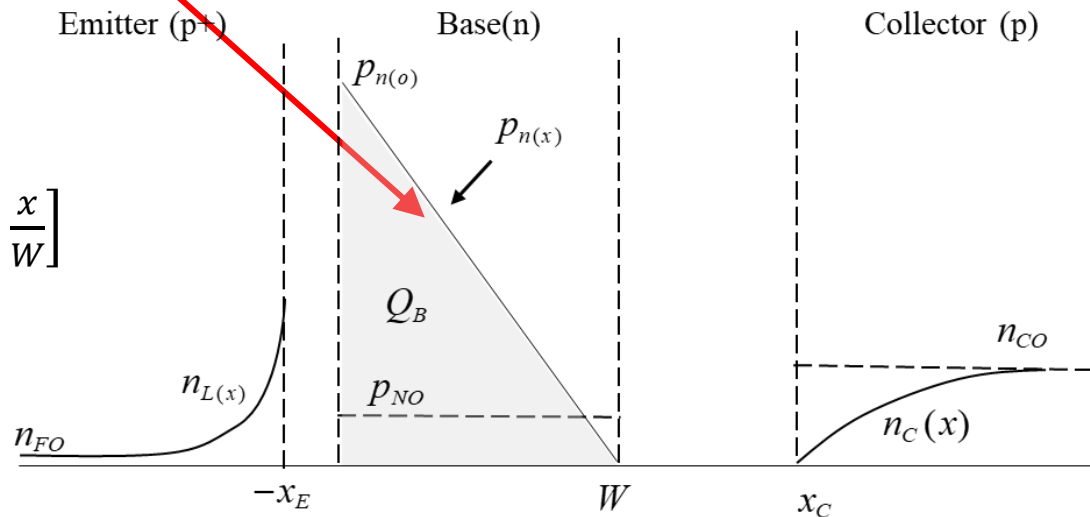


Similarly, the hole current collected by the collector at $x = W$ is

$$I_{Cp} = A \left(-q D_p \frac{dp}{dx} \Big|_{x=W} \right) \approx q A \frac{D_p p_{no}}{W} e^{qV_{EB}/kT} \quad (21)$$

Note that I_{Cp} is equal to I_{Ep} when $W/L_p \ll 1$.

$$p_n(x) = p_{no} e^{qV_{EB}/kT} \left[1 - \frac{x}{W} \right]$$



The electron current I_{En} , which is due to electron flow from the base to the emitter, and I_{Cn} , which is due to the electron flow from the collector to the base, are

$$I_{En} = A \left(q D_{En} \frac{dn_E}{dx} \Big|_{x = -x_E} \right) = q A \frac{D_{En} n_{EO}}{L_E} (e^{qV_{EB}/kT} - 1) \quad (22)$$

$$I_{Cn} = A \left(q D_{Cn} \frac{dn_C}{dx} \Big|_{x = x_C} \right) = q A \frac{D_{Cn} n_{CO}}{L_C} \quad (23)$$

Where D_{En} and D_{Cn} are the diffusion coefficients of electron in the emitter and collector, respectively.

$$\begin{aligned}
 \frac{dn_E}{dx} &= \frac{d}{dx} \left[n_{E0} + \left(n_{E0} e^{qV_{EB}/kT} - n_{E0} \right) e^{\frac{x+x_E}{L_E}} \right] \\
 &= \frac{d}{dx} \left[\cancel{n_{E0}} + n_{E0} e^{qV_{EB}/kT} \cdot e^{\frac{x+x_E}{L_E}} - n_{E0} e^{\frac{x+x_E}{L_E}} \right] \\
 &= n_{E0} \frac{e^{qV_{EB}/kT}}{L_E} \cdot e^{\frac{x+x_E}{L_E}} \Big|_{x=-x_E} - \frac{n_{E0}}{L_E} e^{\frac{x+x_E}{L_E}} \Big|_{x=-x_E} \\
 &= \frac{n_{E0}}{L_E} e^{qV_{EB}/kT} - \frac{n_{E0}}{L_E} = \frac{n_{E0}}{L_E} \left(e^{qV_{EB}/kT} - 1 \right)
 \end{aligned}$$

$$\begin{aligned}
 \frac{dn_C}{dx} &= \frac{d}{dx} \left[\cancel{n_{C0}} - n_{C0} e^{-\frac{x-x_C}{L_C}} \right] \\
 &= - \frac{n_{C0}}{-L_C} e^{-\frac{x-x_C}{L_C}} \Big|_{x=x_C} = \frac{n_{C0}}{L_C} e
 \end{aligned}$$

Terminal Currents

The terminal currents can be obtained from these equations.

The **emitter current** is the sum of Eqs. 20 and 22, i.e., $I_{Ep} + I_{En}$

$$I_E = a_{11}(e^{qV_{EB}/kT} - 1) + a_{12} \quad (24)$$

$$a_{11} \cong qA \left(\frac{D_p p_{no}}{W} + \frac{D_E n_{EO}}{L_E} \right) \quad (25)$$

$$a_{12} \cong \frac{qAD_p p_{no}}{W} \quad (26)$$

The **collector current** is the sum of Eqs. 21 and 23, i.e., $I_{Cp} + I_{Cn}$

$$I_C = a_{21}(e^{qV_{EB}/kT} - 1) + a_{22} \quad (27)$$

$$\text{Where } a_{21} \cong \frac{qAD_p p_{no}}{W} = a_{12} \quad (28)$$

$$a_{22} \cong qA \left(\frac{D_p p_{no}}{W} + \frac{D_{Cn} n_{CO}}{L_C} \right) \quad (29)$$

The **base current** is the difference between the emitter current and collector current,

$$I_B = (a_{11} - a_{21})(e^{qV_{EB}/kT} - 1) + (a_{12} - a_{22}) \quad (30)$$

The currents in the three terminals of a transistor are mainly determined by **the minority carrier distribution in the base region**.

For the case of $W/L_p \ll 1$, we can simplify the emitter efficiency from Eqs. 20 and 22:

$$\gamma = \frac{I_{Ep}}{I_{Ep} + I_{En}} = \frac{\frac{D_p p_{no}}{W}}{\frac{D_p p_{no}}{W} + \frac{D_E n_{EO}}{L_E}} = \frac{1}{1 + \frac{D_E n_{EO} W}{D_p p_{no} L_E}} \quad (31)$$

or

$$\gamma = \frac{1}{1 + \frac{D_E N_B W}{D_p N_E L_E}} \quad (31a)$$

To get the maximum emitter efficiency, $\frac{D_E N_B W}{D_p N_E L_E}$ must be close to zero. Therefore,

- $N_B \ll N_E$ (Emitter must be more heavily doped than Base.)
- $D_E \ll D_p$
- $W \ll L_E$

$$\gamma = \frac{I_{Ep}}{I_{Ep} + I_{En}} = \frac{qA \frac{D_p P_{n0}}{W} e^{qV_{EB}/kT}}{qA \frac{D_p P_{n0}}{W} e^{qV_{EB}/kT} + qA \frac{D_{En} n_{E0}}{L_E} (e^{qV_{EB}/kT} - 1)}$$

(e^{qV_{EB}/kT} - 1) ignored

$$= \frac{D_p P_{n0}/W}{\frac{D_p P_{n0}}{W} + \frac{D_{En} n_{E0}}{L_E}} = \frac{D_p P_{n0}/W}{D_p P_{n0}/W \left(1 + \frac{D_{En} n_{E0}}{D_p P_{n0}} \cdot \frac{W}{L_E} \right)}$$

$$= \frac{1}{1 + \frac{D_E}{D_p} \cdot \frac{n_{E0}}{P_{n0}} \cdot \frac{W}{L_E}} = \frac{1}{1 + \frac{D_E}{D_p} \cdot \frac{N_B}{N_E} \cdot \frac{W}{L_E}}$$

$$n_{E0} = \frac{n_i^2}{N_E}$$

$$P_{n0} = \frac{n_i^2}{N_B}$$

Example 2

An ideal p⁺-n-p transistor has impurity concentrations of 10^{19} , 10^{17} , and $5 \times 10^{15} \text{ cm}^{-3}$ in the emitter, base and collector regions, respectively. The corresponding lifetimes are 10^{-8} , 10^{-7} , and 10^{-6} s . Assume an effective cross section area A is 0.05 mm^2 and the E - B junction is forward biased to 0.6 V . Find the common base current gain of the transistor. Note that the other device parameters are $D_E = 1 \text{ cm}^2/\text{s}$, $D_p = 10 \text{ cm}^2/\text{s}$, $D_C = 1 \text{ cm}^2/\text{s}$, and $W = 0.5 \text{ }\mu\text{m}$.

Solution

In the base region,

$$L_p = \sqrt{D_p \tau_p} = \sqrt{10 \times 10^{-7}} = 10^{-3} \text{ cm}$$

$$p_{no} = n_i^2 / N_B = (9.65 \times 10^9)^2 / 10^{17} = 9.31 \times 10^2 \text{ cm}^{-3}$$

In the emitter, $L_E = \sqrt{D_E \tau_E} = 10^{-4} \text{ cm}$ and $n_{EO} = n_i^2 / N_E = 9.31 \text{ cm}^{-3}$

Since $W/L_p = 0.05 \ll 1$, the current components are given by

$$I_{Ep} = qA \frac{D_p p_{no}}{L_p} e^{qV_{EB}/kT} = \frac{1.6 \times 10^{-19} \times 5 \times 10^{-4} \times 10 \times 9.31 \times 10^2}{0.5 \times 10^{-4}} e^{0.6/0.0259}$$

$$= 1.7137 \times 10^{-4} \text{ A}$$

$$I_{Cp} = 1.7137 \times 10^{-4} \text{ A}$$

$$I_{En} = qA \frac{D_E n_{EO}}{L_E} (e^{qV_{EB}/kT} - 1) = \frac{1.6 \times 10^{-19} \times 5 \times 10^{-4} \times 1 \times 9.31}{10^{-4}} (e^{0.6/0.0259} - 1)$$

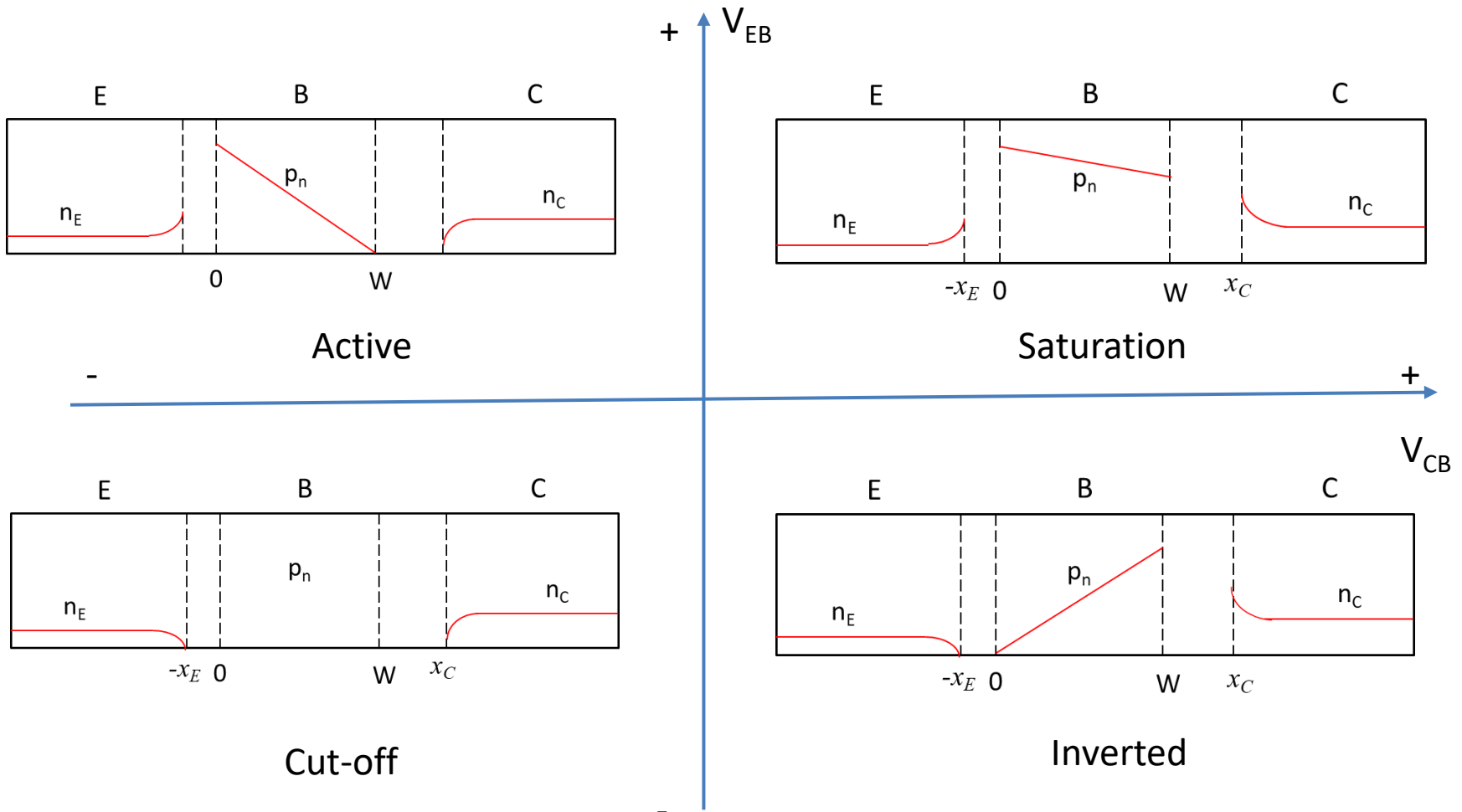
$$= 8.5687 \times 10^{-8} \text{ A}$$

Therefore, the common-base current gain

$$\alpha_o = \frac{I_{Cp}}{I_{Ep} + I_{En}} = \frac{1.7137 \times 10^{-4}}{1.7137 \times 10^{-4} + 8.5687 \times 10^{-8}} = 0.9995$$

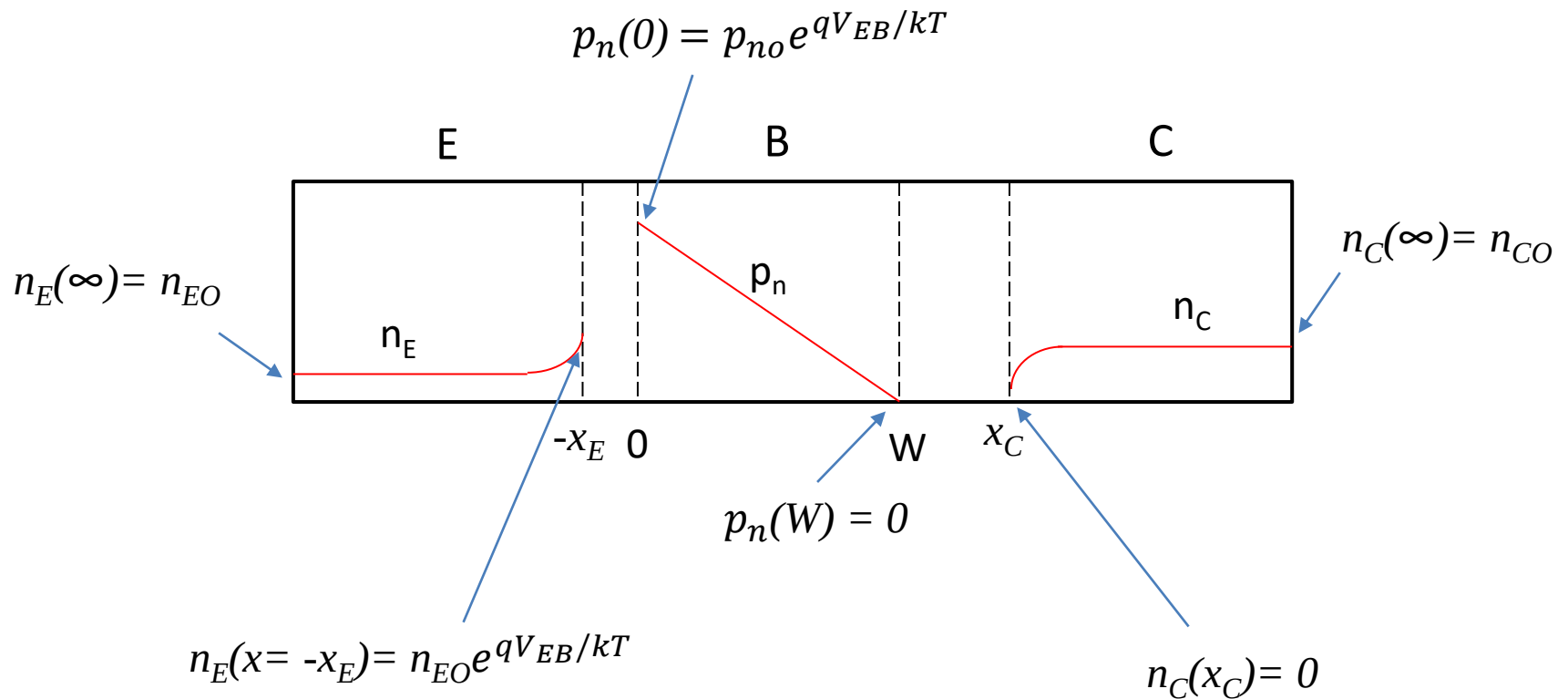
2.3 Mode of Operation

A bipolar transistor has four modes of operation, depending on the voltage polarities on the E - B junction and the B - C junction.



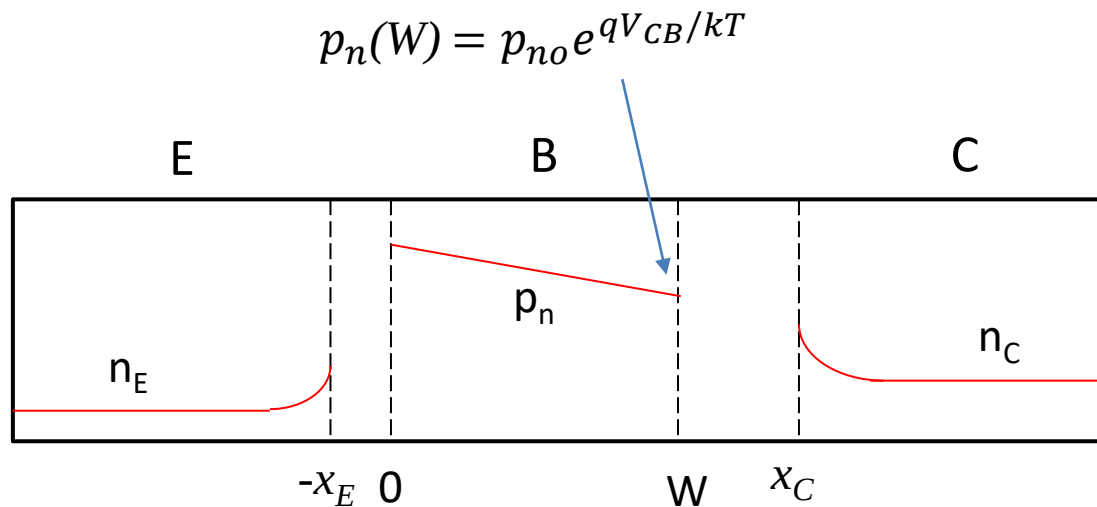
(1) Active Mode

The E - B junction is forward-biased and B - C junction is reverse-biased. The minority carrier distributions are as shown below.



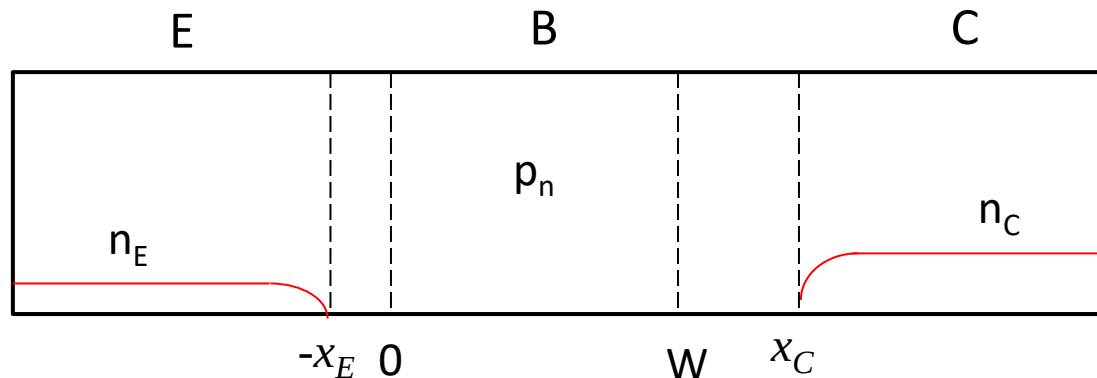
(2) Saturation mode

Both junctions are forward-biased. The boundary condition at $x = W$ becomes $p_n(W) = p_{n0}e^{qV_{CB}/kT}$ instead of zero. The saturation mode corresponds to small biasing voltage and large output current, that is, the transistor is in conducting state and acts as a **closed (or on) switch**.



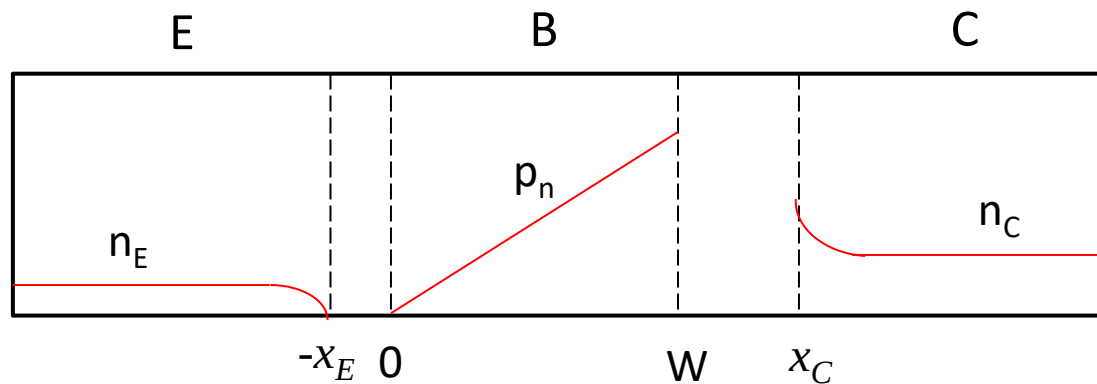
(3) Cutoff mode

Both junctions are reverse-biased. The boundary conditions become $p_n(0) = p_n(W)=0$. The cutoff mode corresponds to the **open (or off) state** of the transistor as a switch.



(4) Inverted mode

It is sometime called the inverted active mode. In the mode, the $E-B$ junction is reverse-biased and the $C-B$ junction is forward-biased. The inverted mode corresponds to the case where the collector acts like an emitter and the emitter acts like a collector, that is, the device is used backward.



The current gain for the inverted mode is generally lower than that for the active mode because of poor “emitter efficiency” resulting from low collector doping with respect to the base doping.

The current-voltage relationships for the various modes of operation can be obtained by following the same procedures used for the active mode, with appropriate changes of the boundary conditions. The general expression applicable to all modes of operations are

$$I_E = a_{11}(e^{qV_{EB}/kT} - 1) - a_{12}(e^{qV_{CB}/kT} - 1) \quad (32a)$$

$$I_C = a_{21}(e^{qV_{EB}/kT} - 1) - a_{22}(e^{qV_{CB}/kT} - 1) \quad (32b)$$

Where the coefficients a_{11} , a_{12} , a_{21} and a_{22} are the same as given before.

The biasing voltages for the junctions can be positive or negative depending on the mode of operation.

2.4 Current-voltage characteristics of common-base and common-emitter configuration

Common-base configuration

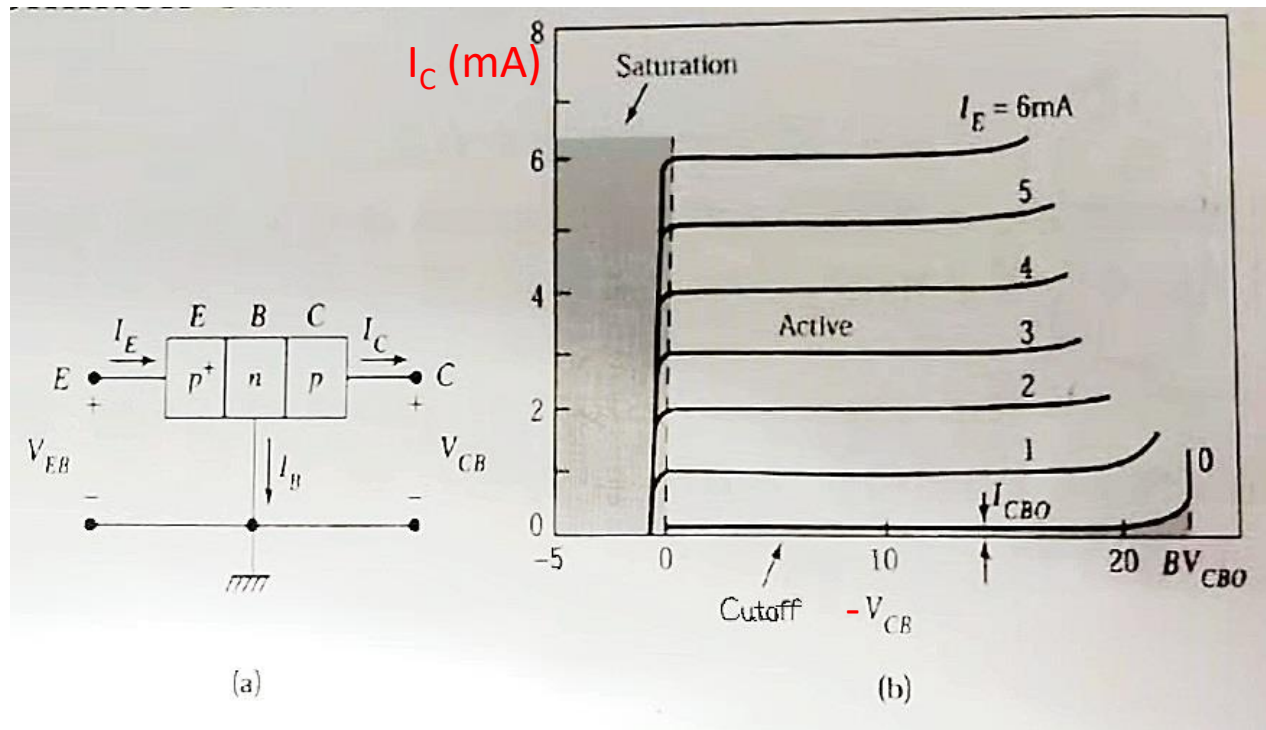


Fig.8 (a) Common-base configuration of a $p-n-p$ transistor.
(b) Its output current-voltage characteristics.

Common-emitter configuration

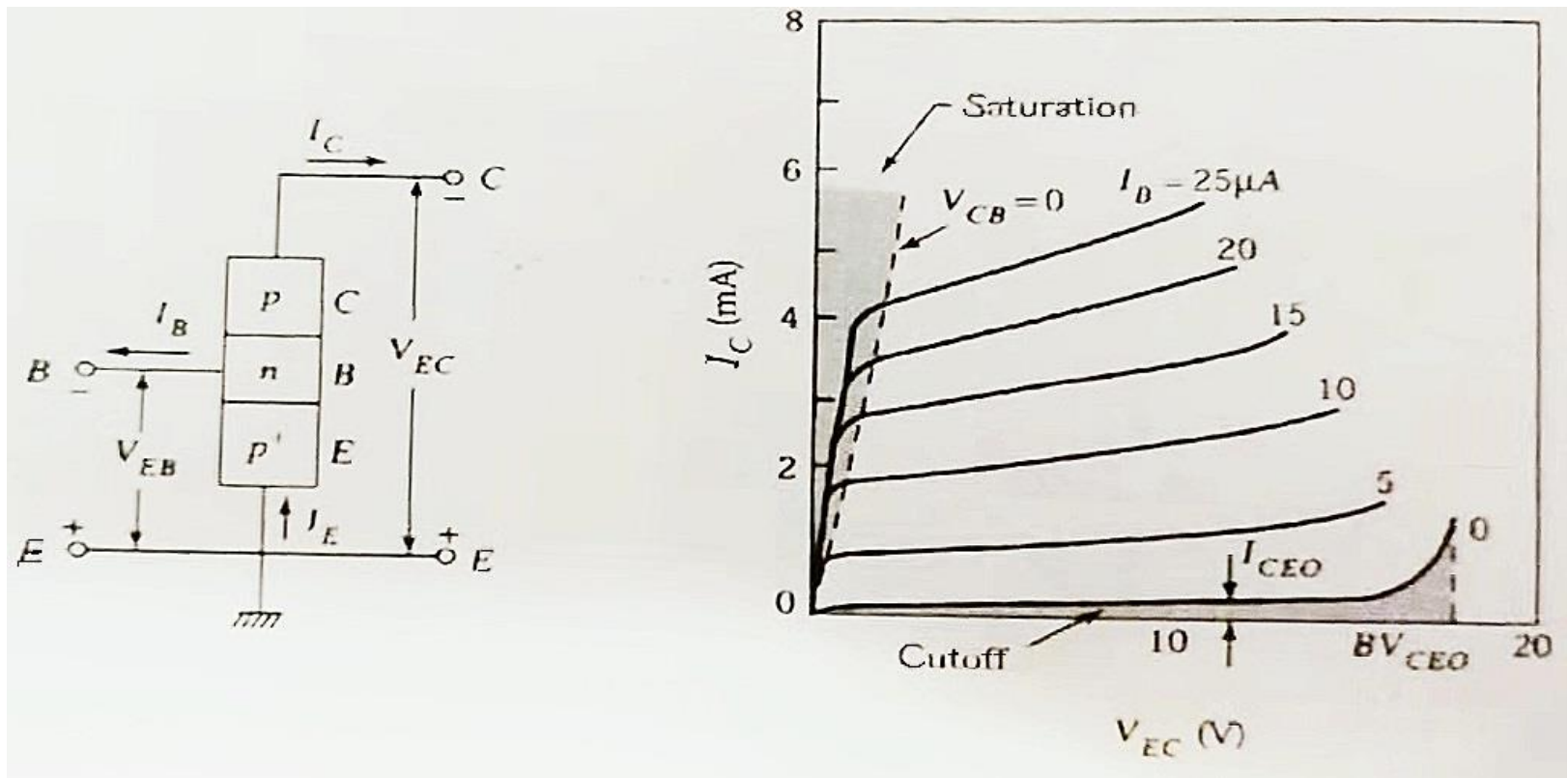


Fig.10 (a) Common-emitter configuration of a *p-n-p* transistor.
(b) Its output current-voltage characteristics.

The collector current for common-emitter

From $I_C = I_{Cp} + I_{Cn}$ and $I_C = \alpha_o I_E + I_{CBO}$

We get

$$I_C = \alpha_o (I_B + I_C) + I_{CBO} \quad (33)$$

Solving for I_C ,

$$I_C = \frac{\alpha_o}{1 - \alpha_o} I_B + \frac{I_{CBO}}{1 - \alpha_o} \quad (34)$$

We designate β_0 as the *common-emitter current gain*, which is the incremental change with respect to an incremental change of I_B . From Eq. 34 we have

$$\beta_0 = \frac{\Delta I_C}{\Delta I_B} = \frac{\alpha_o}{1 - \alpha_o} \quad (35)$$

We also designate I_{CEO} as

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha_o} \quad (36)$$

Which corresponds to the collector-emitter leakage current for $I_B = 0$. Eq.34 becomes

$$I_C = \beta_0 I_B + I_{CEO} \quad (37)$$

Usually, the α_o is close to unity, so β_0 is much larger than 1.

Example 3

Referring to Example 1. For an ideal $p-n-p$ transistor, the current components are given by $I_{Ep}=3$ mA, $I_{En}=0.01$ mA, $I_{Cp}=2.99$ mA and $I_{Cn}= 0.001$ mA. Find the common-emitter current gain β_0 . Express I_{CEO} in terms of β_0 and I_{CBO} and find the value of I_{CEO} .

Solution

The α_o in Example 1 is 0.9934. Hence

$$\beta_0 = \frac{\alpha_o}{1-\alpha_o} = \frac{0.9934}{1-0.9934} = 150.5$$

Eq.36 can be expressed by

$$I_{CEO} = \frac{I_{CBO}}{1-\alpha_o} = \left(\frac{\alpha_o}{1-\alpha_o} + 1\right) I_{CBO} = (\beta_0 + 1) I_{CBO}$$

Therefore,

$$I_{CEO} = (\beta_0 + 1) I_{CBO} = (150.5 + 1) \times 0.87 \times 10^{-6}$$

$$= 1.32 \times 10^{-4} \text{ A}$$

3. Frequency Response of Bipolar Transistors

3.1 The *Ebers-Moll* model

We shall discuss the *Ebers-Moll model*, which is applicable for all modes of operation, and then extend the mode to discuss the frequency response.

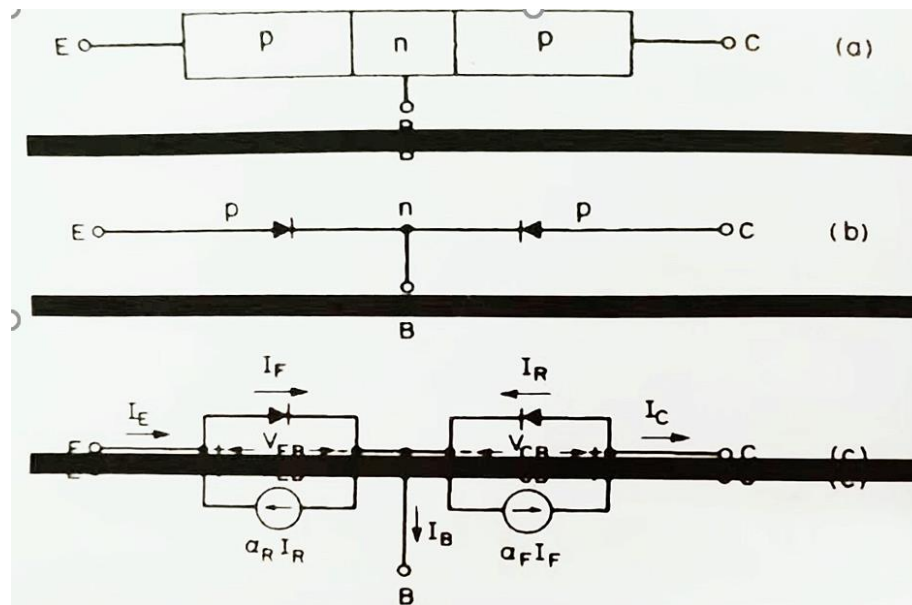
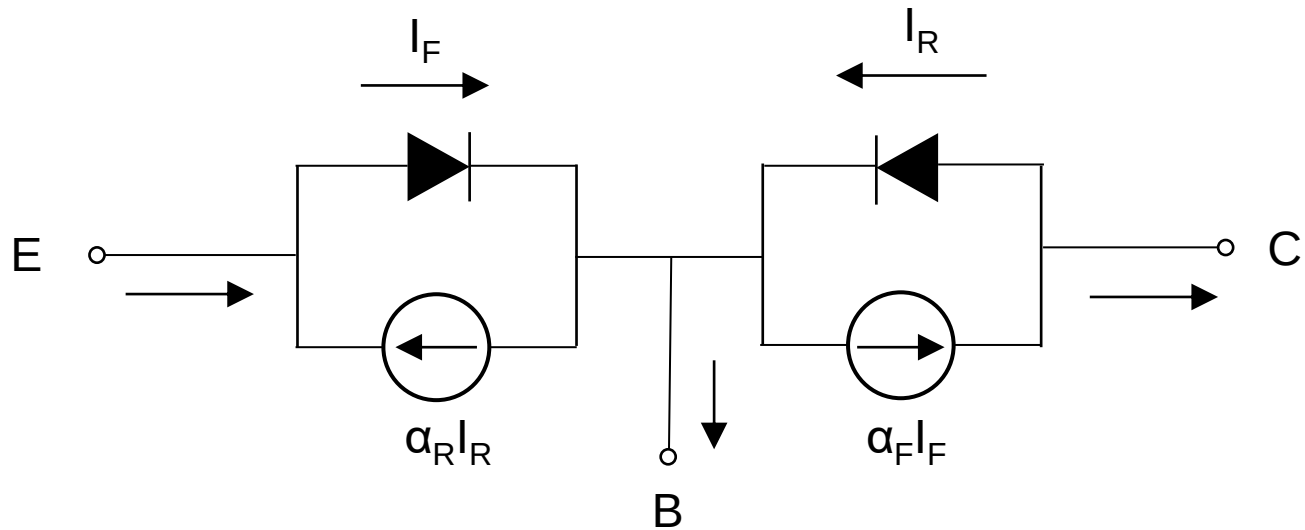
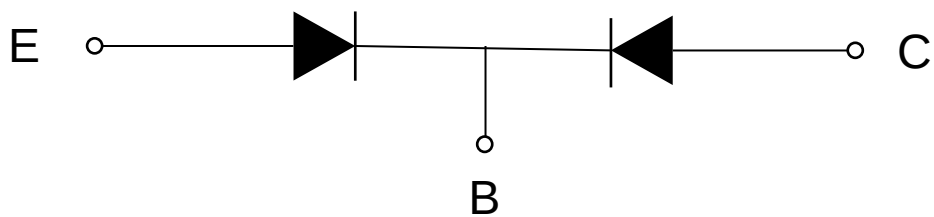
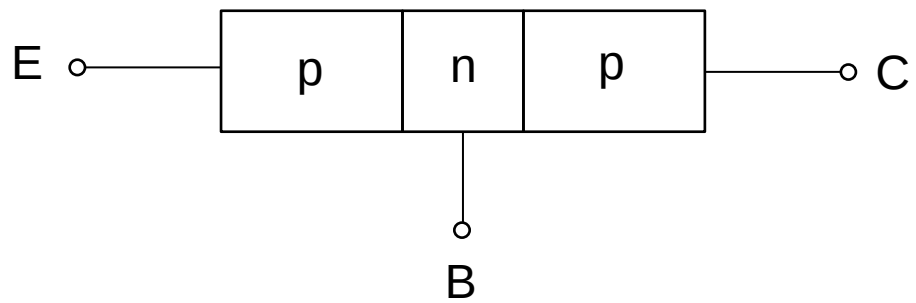
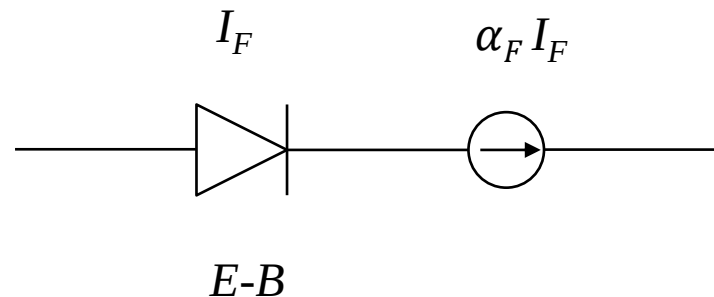


Fig.13 (a) Cross section of a p-n-p transistor. (b) A p-n-p transistor represented as two p-n junction diodes connected back-to-back. (c) Circuit diagram of *Ebers-Moll* model.



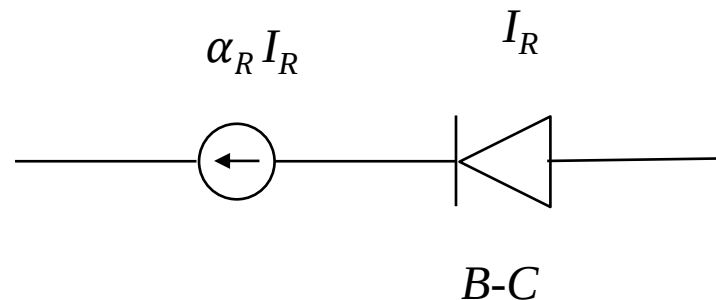
In the *Ebers-Moll* model, the current crossing the emitter-base junction is shown as I_F .

In the active mode, a large fraction of the minority carrier injected into the base from the emitter will reach the collector. The fraction is represented by the current generator $\alpha_F I_F$, where α_F is the common-base current gain.



In the inverted mode of operation, the collector-base junction is forward biased and the emitter-base junction is reverse-biased. This mode is represented by the collector diode current I_R , which is a function of the collector-base voltage V_{CB} , and the current generator $\alpha_R I_R$ across the emitter-base junction.

The factor α_R is called the *reverse common-base current gain* and is the ratio of the number of minority carriers collected at the emitter to the number of minority carriers injected into the base at the forward-biased collector-base junction.



The equations for the diode current I_F and I_R are:

$$I_F = I_{FO} (e^{qV_{EB}/kT} - 1) \quad (39)$$

$$I_R = I_{RO} (e^{qV_{CB}/kT} - 1) \quad (40)$$

Where I_{FO} and I_{RO} are the saturation currents of the E - B and C - B p-n junctions, respectively.

The terminal currents are

$$I_E = I_F - \alpha_R I_R \quad (41)$$

$$I_C = \alpha_F I_F - I_R \quad (42)$$

$$I_B = I_E - I_C = (1 - \alpha_F) I_F + (1 - \alpha_R) I_R \quad (43)$$

By combining Eqs. 39 to 42, we obtain the general equations for the basic *Ebers-Moll* model:

$$I_E = I_{FO} (e^{qV_{EB}/kT} - 1) - \alpha_R I_{RO} (e^{qV_{CB}/kT} - 1) \quad (44)$$

$$I_C = \alpha_F I_{FO} (e^{qV_{EB}/kT} - 1) - I_{RO} (e^{qV_{CB}/kT} - 1) \quad (45)$$

The *Ebers-Moll* model has four parameters: I_{FO} , I_{RO} , α_F and α_R . Comparing Eqs. 44 and 45 with Eq. 32 gives

$$I_{FO} = a_{11} \quad (46a) \quad \alpha_R I_{RO} = a_{12} \quad (46b)$$

$$\alpha_F I_{FO} = a_{21} \quad (46c) \quad I_{RO} = a_{22} \quad (46d)$$

Where a_{11} , a_{12} , a_{21} and a_{22} are defined previously.

Because $a_{12} = a_{21}$, therefore, $\alpha_R I_{RO} = \alpha_F I_{FO}$. Thus, only three parameters (e.g., I_{FO} , I_{RO} and α_F) are required for the basic *Ebers-Moll* model.

3.2 Frequency Response

- In the previous discussions, we were concerned with the static (or dc) characteristics of the bipolar transistor.
- We shall now study its ac characteristics when a small-signal voltage or current superimposed upon the dc value.
- The term “small-signal” means that the peak values of the ac signal current and voltage are smaller than the dc values.

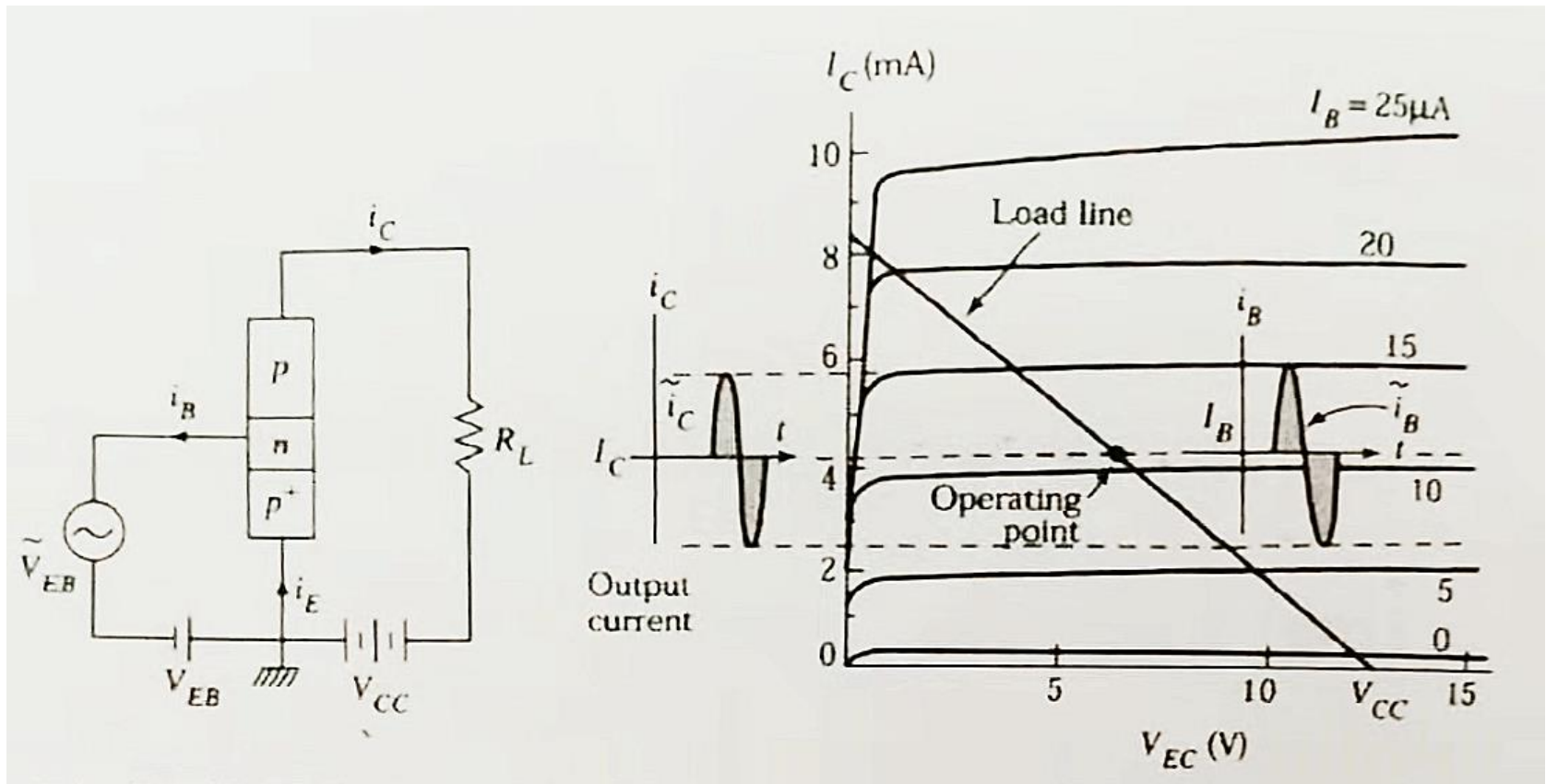


Fig. 14 (a) Bipolar transistor connected in the common-emitter configuration. (b) Small signal operation of the transistor.

For a dc bias voltage V_{EB} , a dc base current I_B and dc collect current I_C flow in the transistor. These currents correspond to the operating point as shown in Fig. 14b.

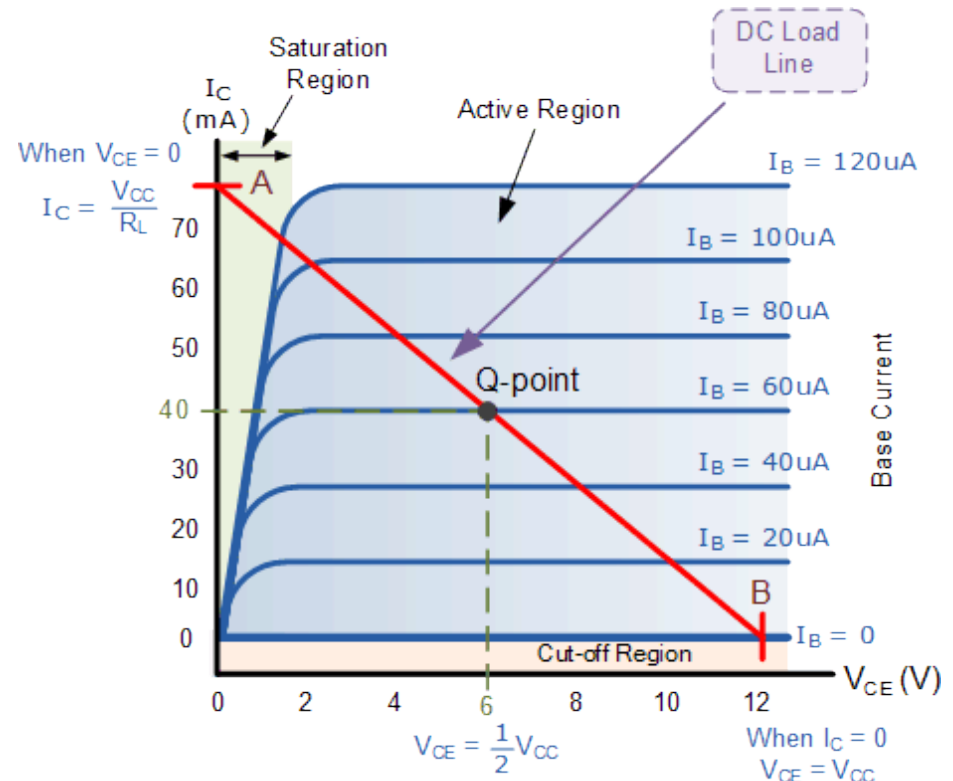
The load line, determined by the supply voltage V_{CC} and the load resistance R_L , intercepts the V_{EC} at V_{CC} and has a slope of $1/R_L$.

$$V_{CC} = V_{CE} + I_C R_L$$

$$I_C = (V_{CC} - V_{CE})/R_L$$

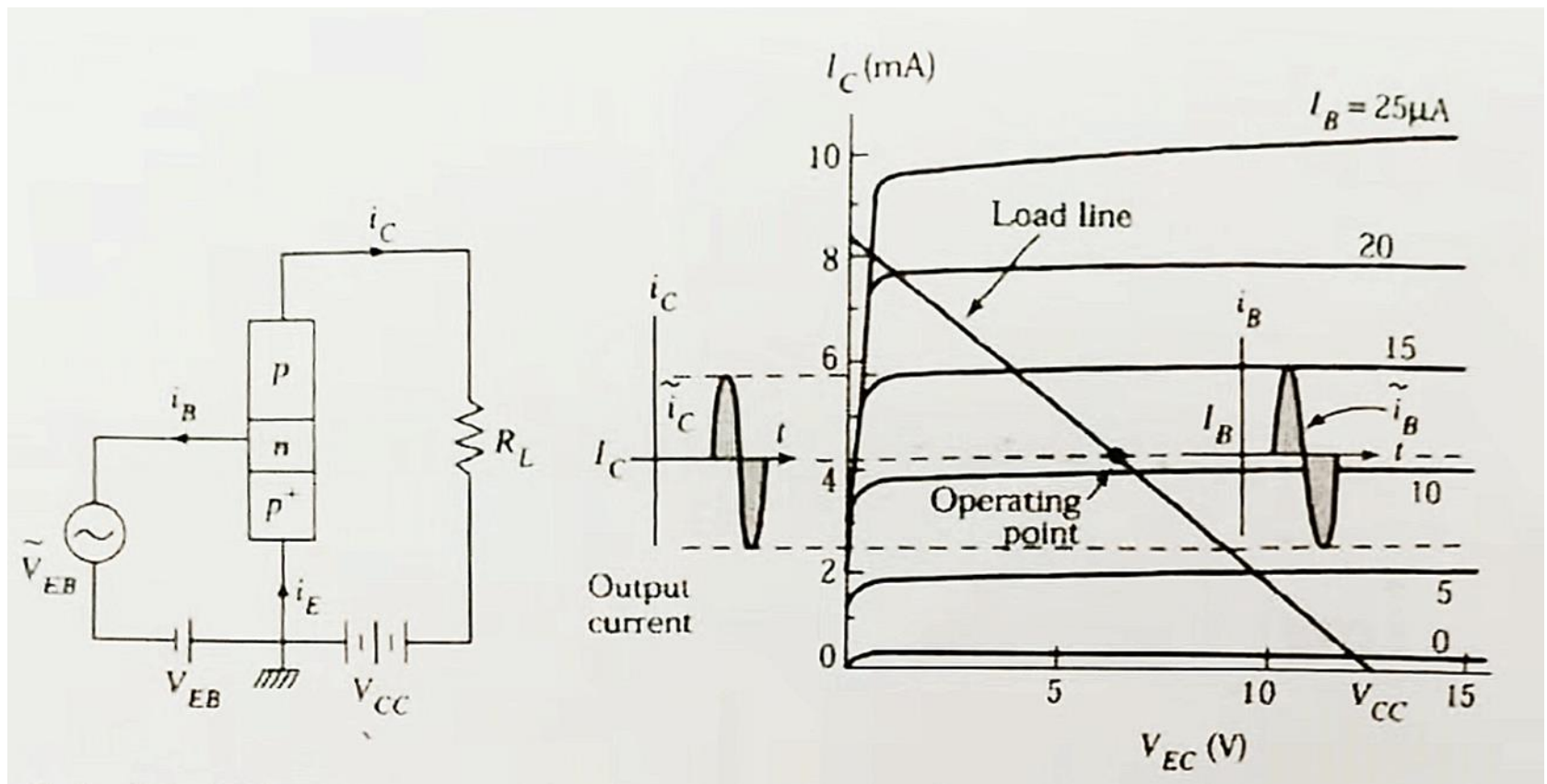
$$I_C = V_{CC}/R_L \text{ when } V_{CE} = 0$$

$$I_C = 0 \text{ when } V_{CE} = V_{CC}$$



When a small ac signal is superimposed on the dc V_{EB} , the base current i_B will vary as a function of time.

This variation, in turn, brings a corresponding variation in the output current i_C , which is β times larger than the input current variation. Thus, the transistor amplifies the input signal.



The total currents and the voltage could be defined as the sum of their dc and ac components:

$$v_{EB} = V_{EB} + v_{eb} \quad (47a)$$

$$i_B = I_B + i_b \quad (47b)$$

$$i_C = I_C + i_c \quad (47c)$$

$$i_E = I_E + i_e \quad (47d)$$

Total = dc + ac

Using the *Ebers-Moll* model and $\exp (qV_{EB}/kT) \gg 1$ and $\exp (qV_{CB}/kT) \ll 1$ in the active mode, we have

$$i_E = I_{FO} (e^{qv_{EB}/kT} - 1) - \alpha_R I_{RO} (e^{qv_{CB}/kT} - 1) \quad (48)$$

$$\approx I_{FO} e^{qv_{EB}/kT}$$

$$i_C = \alpha_F I_{FO} (e^{qv_{EB}/kT} - 1) - I_{RO} (e^{qv_{CB}/kT} - 1) \quad (49)$$

$$\approx \alpha_F I_{FO} e^{qv_{EB}/kT}$$

$$i_B = i_E - i_C \approx (1 - \alpha_F) I_{FO} e^{qv_{EB}/kT} \quad (50)$$

The current i_C and i_B can also be expressed by the Taylor series expansion up to the linear terms of the signal voltage:

$$i_C = I_C + \left. \frac{\partial i_C}{\partial V_{EB}} \right|_{V_{EC}} v_{eb} = I_C + g_m v_{eb} = I_C + i_c \quad (51)$$

Where g_m is called the *transconductance*

$$g_m = \left. \frac{\partial i_C}{\partial V_{EB}} \right|_{V_{EC}} = \alpha_F I_{FO} \left(\frac{q}{kT} \right) e^{qV_{EB}/kT} \quad (52)$$

$$= \alpha_F \frac{qI_E}{kT} = \frac{qI_C}{kT} = \frac{I_C}{V_T}$$

For i_B

$$i_B = I_B + \left. \frac{\partial i_B}{\partial V_{EB}} \right|_{V_{EC}} v_{eb} = I_B + g_{EB} v_{eb} = I_B + i_b \quad (53)$$

g_{EB} is called the **input conductance**.

$$g_{EB} = \left. \frac{\partial i_B}{\partial V_{EB}} \right|_{V_{EC}} = \frac{q(1-\alpha_F)}{kT} I_E = \frac{1-\alpha_F}{V_T} I_E \quad (54)$$

The transconductance g_m and the input conductance g_{EB} are dependent on the common-base current gain.

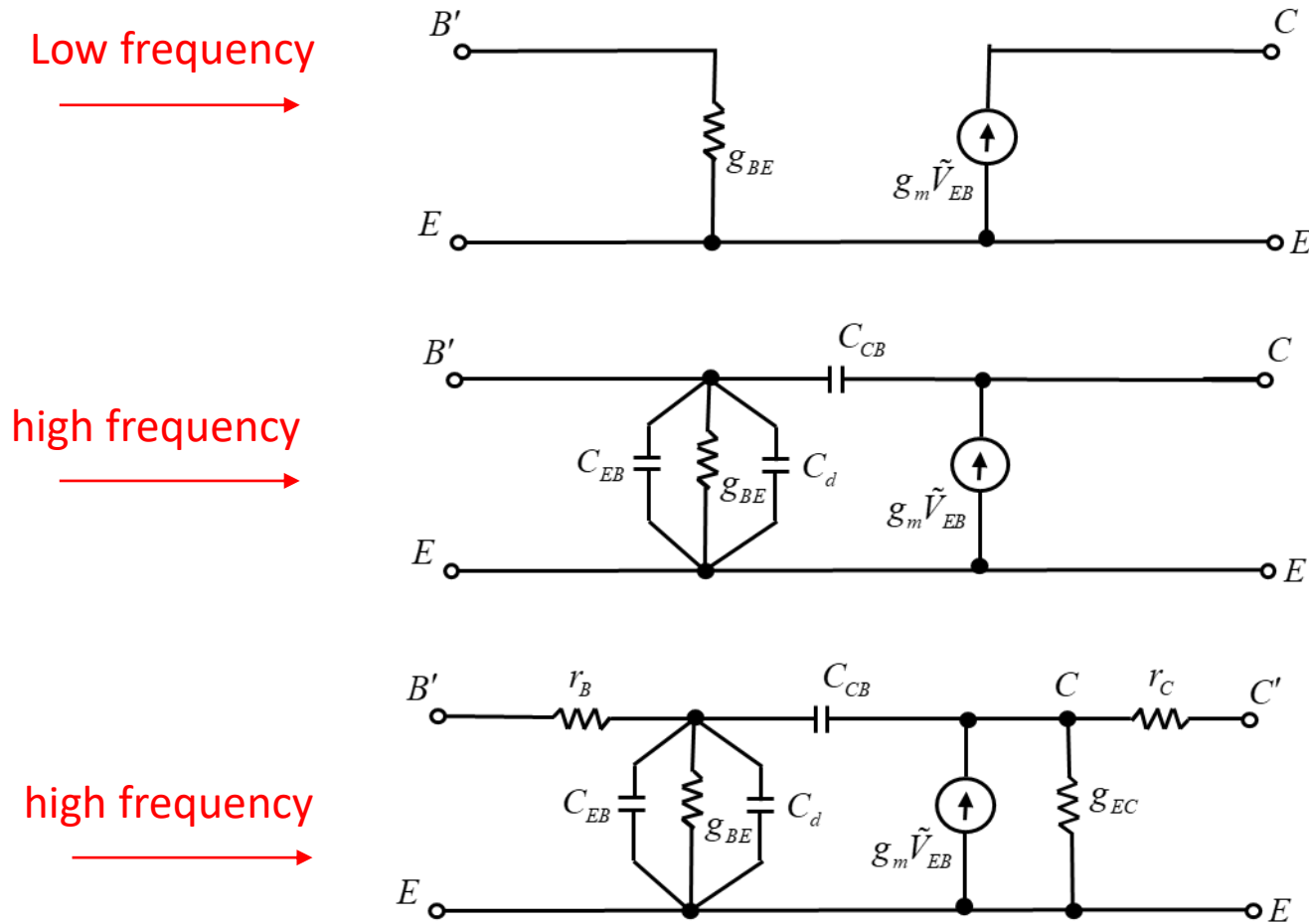


Fig. 15. (a) Basic transistor equivalent circuit.
 (b) Basic circuit with the addition of depletion and diffusion capacitances.
 (c) Basic circuit with the addition of resistance and conductance

Fig. 15a. Shows the equivalent circuit for the low frequency situation, which is called **the hybrid- π model**.

The forward-biased emitter-base junction has a depletion capacitance C_{EB} and a diffusion capacitance C_d . The reverse-biased collector-base junction has a depletion capacitance C_{CB} . **At high frequency situation, these capacitances should be considered**, as shown in Fig.15b.

To account for the base width effect, there is a finite output conductance $g_{EC} = i_c / v_{ec}$ when $v_{eb} = 0$. In addition, we have a base resistance r_B , a collector resistance r_C . Fig.15c is the high frequency equivalent circuit incorporating all elements.

Cut-off frequency

At low frequencies, the current gain is a constant. However, it will decrease after a critical frequency is reached.

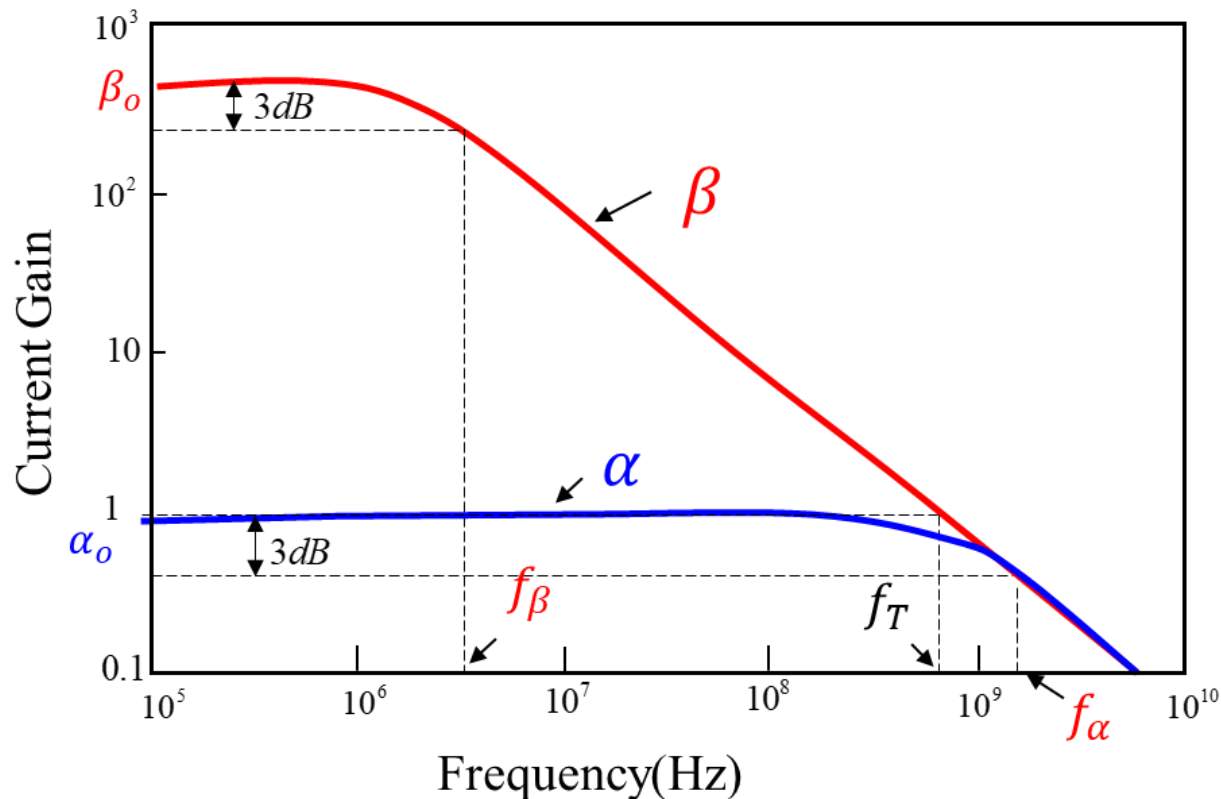


Fig. 16. Current gain as a function of operation frequency.

Common-base current gain

The common-base current gain α can be described as

$$\alpha = \frac{\alpha_0}{1 + j(f/f_\alpha)} \quad (55)$$

Where α_0 is the low frequency (or dc) common-base current gain and f_α is the common-base cut-off frequency.

At $f = f_\alpha$, the magnitude of α is $0.707 \alpha_0$ (3 dB down).

Common-emitter current gain

The common-emitter current gain β can be written as

$$\beta = \frac{\beta_0}{1 + j(f/f_\beta)} \quad (56)$$

Where f_β is the common-emitter cutoff frequency and is given by

$$f_\beta = (1 - \alpha_0) f_\alpha \quad (57)$$

Since $\alpha_0 \approx 1$, f_β is much smaller than f_α .

Cut-off frequency f_T

The cut-off frequency is f_T the frequency when $|\beta|$ becomes unity.

By setting

$$\beta = \frac{\beta_0}{1+j(f/f_\beta)} = \frac{\beta_0}{1+j(f_T/f_\beta)} = 1$$

we obtain

$$f_T = \sqrt{\beta_0^2 - 1} f_\beta \cong \beta_0(1 - \alpha_0) f_\alpha \cong \alpha_0 f_\alpha \quad (58)$$

f_T is very close to but is smaller than f_α .

f_T - τ_B Relationship

The cut-off frequency f_T can also be expressed as $(2\pi\tau_T)^{-1}$ where τ_T is the total time of the carrier transit from the emitter to the collector.

τ_T includes the emitter delay time τ_E , the base transit time τ_B , and the collector transit time τ_C .

The most important delay time is τ_B .

The distance travelled by the minority carrier in the base in a time interval dt is $dx=v(x)dt$, where $v(x)$ the effective minority-carrier velocity in the base.

The effective minority-carrier velocity in the base is related to the current as

$$I_p = qv(x)p(x)A \quad (59)$$

Where A is the device area and $p(x)$ is the distribution of the minority carriers.

The transit time τ_B required for a hole to traverse the base is given by

$$\tau_B = \int_0^W \frac{dx}{v(x)} = \int_0^W \frac{qp(x)A}{I_p} dx \quad (60)$$

For a straight-line hole distribution, as given in Eq. 15, the integration of Eq. 60, using Eq. 21(I_{Cp}) for I_p , leads to

$$\tau_B = \frac{W^2}{2D_p} \quad (61)$$

So

$$f_T = \frac{1}{2\pi\tau_B} = \frac{2D_p}{2\pi W^2} = \frac{D_p}{\pi W^2} \quad (62)$$

Strategy to get high f_T

1. Design transistor with **small base width**.
2. Use **n-p-n type transistor** because the electron diffusion constant in Si is about three times larger than that of holes,.
3. Use **a graded base with a built-in electric field**. For a large doping variation (high doping near the emitter and low doping near the collector), the built-in field in the base will aid the motion of carriers toward the collector and reduce base transit time.

Summary

The bipolar transistor, developed in 1947, remain one of the most important semiconductor devices. A bipolar transistor is formed when two p-n junctions of the same semiconductor materials are physically close enough to interact. Charge carriers injected from the forward-biased first junction result in a large current flow in the reverse-biased second junction.

We have considered the static characteristics of bipolar transistors, such as the modes of operation and the current-voltage characteristics of the common-emitter configuration, as well as the frequency response. *A key device parameter of bipolar transistor is the base width, which must be very small compared with the minority-carrier diffusion length to improve the current gain and to increase the cutoff frequency.*

Bipolar transistors are used extensively as discrete devices or in integrated circuits for current-gain, voltage-gain, and power-gain applications. They are also used in bipolar CMOS combination circuits (BiCMOS), covered for high-density, high-speed operations.